



University of Michigan
Department of Naval Architecture and
Marine Engineering
Conceptual Ship Design Software
Environment

Welcome to the University of Michigan software environment for the conceptual design of displacement type ships. This collection of software has been developed over the years to support the teaching of conceptual ship design within the undergraduate curriculum of the Department of Naval Architecture and Marine Engineering. The emphasis has primarily been on analysis methods that can be utilized at the parameter stage of design before a detailed hull design is developed and offsets are available. The legacy FORTRAN programs were developed by and under the supervision of Professor Michael G. Parsons over the period 1983-1995 and used as Macintosh applications in the senior design courses NA470 Ship Design and NA475 Ship Design Project. During 1997-1998, Dr. Jun Li developed Visual C++ Windows Graphical User Interfaces for the programs as part of the DARPA sponsored COMPASS project. These Windows programs are now available through the World Wide Web. The individuals involved, the department, and the university can, however, accept no responsibility for problems resulting from the use of these programs.

OVERVIEW. The following programs are available:

- Power Prediction Program (PPP) calculates the resistance and hull/propeller interaction of a displacement hull using the regression-based method of Holtrop and Mennen. This uses a modified Hughes Method. An explicit air drag model has been added.
- Propeller Optimization Program (POP) calculates the optimum open water efficiency Wageningen B-Screw Series propeller subject to user-specified diameter and cavitation constraints. The nonlinear programming Nelder and Mead Simplex Search method is used with an External Penalty Function. The propeller RPM can either be free or specified. The program will also calculate the operating conditions and efficiency for a specified operating point.
- Maneuvering Prediction Program (MPP) uses (1) methods presented by Clarke et al to assess the course stability, turnability, and controllability by a helmsman, and (2) the regression results of Lyster and Knights to estimate the turning circle characteristics of a displacement vessel.
- Seakeeping Prediction Program (SPP) is an adaptation of the five degree-of-freedom SCORES program to provide estimates of heave, pitch, roll, vertical wave bending moment, horizontal wave bending moment, and hull torsional

moment in a random, long-crested seaway. It can be used at the parameter stage for initial estimates or, with greater accuracy, after the specific Section Area Curve and Design Waterline Curves are available. Hull offsets are not required.

- Gear Sizing Program (GSP) adapts the method presented by Balukjian to provide initial estimates of the size and arrangement of a marine reduction gear for a given input, output, and K-factor gear tooth contact stress loading.

The methods, input, and output of each of these programs will be described in succession.

Note that for these programs to run successfully, the entire program folder as provided must be located in a region of the computer to which the user has write privileges.



POWER PREDICTION PROGRAM (PPP1.8)

M. G. Parsons, January, 1996

REFERENCES:

1. Holtrop, J., and Mennen, G. G. J., "An Approximate Power Prediction Method," *International Shipbuilding Progress*, Vol. 29, No. 335, July, 1982.
2. Holtrop, J., "A Statistical Re-Analysis of Resistance and Propulsion Data," *International Shipbuilding Progress*, Vol. 31, No. 363, November, 1984.

METHODS:

This program implements the statistical ship power estimation method presented by Holtrop and Mennen and Holtrop (1, 2). This model is based upon hydrodynamic theory with coefficients obtained from the regression analysis of the results of 334 ship model tests conducted at MARIN. The user should usually consult these papers prior to using the program. An explicit air drag estimate has been added. The range of applicability of this estimation method is stated to be the following:

$$\begin{aligned}0.55 &\leq C_p \leq 0.85 \\3.90 &\leq L/B \leq 14.9 \\2.10 &\leq B/T \leq 4.00 \\0.05 &\leq F_n \leq 1.00\end{aligned}$$

The program requires the completion of all of the **Input** menu windows and then it can be run from either the **Analysis** menu or the **RunPPP** button.

For simplicity, the program automatically calculates all the output for eight user-specified speeds. The user inputs an initial speed V_0 in knots and a speed increment V_{INC} in knots. These are used to determine the eight speeds as follows:

$$V_i = V_0 + (i - 1) * V_{INC} \quad , \quad i = 1, 2, \dots, 8.$$

The output from the calculations consists of three tables of information for the eight speeds V_i . The first table gives the speeds in various units, the Froude number, the speed-length ratio, and the traditional resistance coefficients C_F , C_R , and C_A . The components of resistance listed below are then presented in Newtons. The third table gives the total resistance, effective power, and the total required thrust (each including the design margin) and the propulsion factors. The resistance model is Holtrop and Mennen's modified Hughes Method with an explicit air drag estimate and design margin added as follows:

$$R_T = [R_F (1 + K_1) + R_W + R_{APP} + R_B + R_{TR} + R_A + R_{AIR}] (1.0 + DMAR/100)$$

R_F	1957 ITTC frictional resistance
$R_F \cdot K_1$	viscous form resistance
R_{APP}	appendage resistance (1)
R_W	wave-making and wave-breaking resistance
R_B	additional pressure resistance of bulbous bow near surface (1)
R_{TR}	additional pressure resistance of immersed transom (1)
R_A	model-ship correlation resistance
R_{AIR}	air drag assuming $C_D=0.4$ for the hull and $C_D=0.8$ for the deck house/deck cargo
DMAR	design margin on resistance in percent

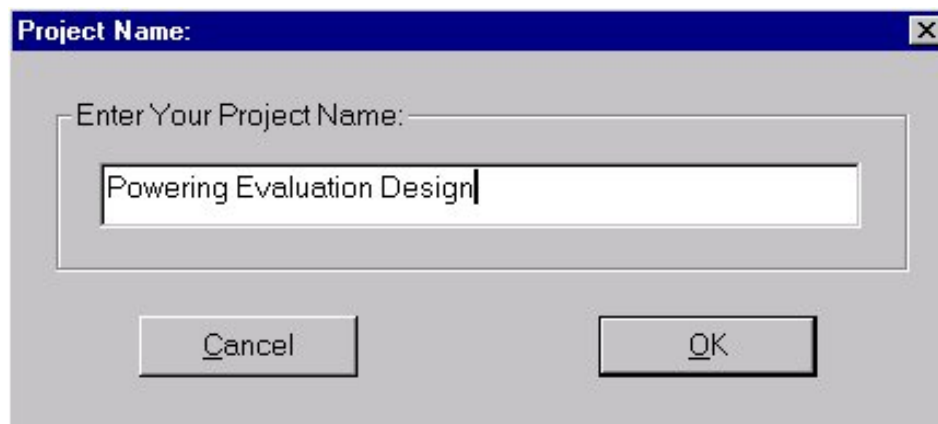
INPUT:

The input to PPP is through a series of eight windows within the menu **Input** as follows:

Input
Project Name
Hull Characteristics
Hull Features
Propulsion Factors Model
Appendages
Modeling
Design Margin
Water Properties

Each of these windows must be completed before an analysis can be performed. The completed input data can be saved using the **Save** button for later re-use.

Project Name provides a location for the general identification of the project being analyzed.



The image shows a standard Windows-style dialog box titled "Project Name:". The main area contains a label "Enter Your Project Name:" and a text box with the text "Powering Evaluation Design" entered. Below the text box are two buttons: "Cancel" and "OK".

Hull Characteristics obtains the basic dimensions and form coefficients for the hull.

Hull Characteristics

Geometry:

Length on waterline (LWL): 205 meters

Maximum beam on LWL (B): 32 meters

Draft forward (TF): 10 meters

Draft aft (TA): 10 meters

Coefficients:

Block coefficient on LWL (CB): 0.5716

Midship coefficient to LWL (CM): 0.98

Waterplane coefficient on LWL (CWP): 0.75

Longitudinal center of buoyancy (LCB): -0.75 %
(in % LWL from amidships; + forward)

Cancel OK

LWL	length on waterline in meters
B	maximum beam on waterline in meters
T _F	draft forward in meters
T _A	draft aft in meters
C _B	block coefficient to LWL
C _M	midship coefficient to LWL
C _{WP}	waterplane coefficient on LWL
LCB	longitudinal center of buoyancy in %LWL from amidships (taken as LWL/2); positive forward

Hull Features obtains information on the bulbous bow, immersed transom, and the shape of the stern.

Hull Features

Bulb:

Transverse bulb area at Station 0 (ABT): 20 meters²

Vertical center of bulb area at Station 0 (HB): 4 meters

(It is recommended that HB ≤ 0.6TF, where TF = draft forward)

Immersed transverse area of transom at zero speed (ATR): 16 meters²

Stern Shape:

☐ Pram with gondola

☐ V-shaped sections

☒ Normally shaped sections

☐ U-shaped sections with Hogner stern

OK Cancel

A_{BT}	transverse bulb area at Station 0 in meters ²
H_B	height of centroid of A_{BT} above baseline in meters (It is recommended that H_B not be allowed to exceed $0.6 \cdot T_F$)
A_{TR}	immersed transverse area of transom at zero speed in meters ²
C_{STERN}	after body shape selected (only $C_{STERN} = 1$ implemented): • pram with gondola (buttock flow stern with wide skeg) • V-shaped sections • normal shaped sections • bulbous U-shaped sections; e.g., Hogner stern

Propulsion Factors Model allows the choice from among three models for the estimation of the hull/propeller interaction based upon the propulsion type:

Propulsion Factors Model

Propulsion Type:

☒ Single-screw with "conventional" stern (SSC)

☐ Single-screw with "open" flow stern (SSOF)

☐ Twin screw (TS)

OK

Cancel

You Selected SSC, Please Enter:

Propeller diameter (D_p): 8 meters

Propeller expanded area ratio (A_e/A_o): 0.7393

Propeller pitch diameter ratio (P/D_p): 0

The required inputs then vary based upon the model selected and these may include:

D_p	propeller diameter in meters; needed only if conventional single screw or twin screw
A_e/A_o	propeller expanded area ratio A_e/A_o ; needed only if conventional single screw
P/D_p	propeller pitch diameter ratio; needed only if twin screw

Appendages allows the selection between two appendage drag models: a simple percent of bare hull resistance or a rational method requiring the wetted area $SAPP_i$ (e.g., both sides of foils) and factor $(1 + K_{2i})$ for each appendage included in the design. The window for the latter case appears as follows:

Appendage	SAPP[i] (meters ²)	(1 + K2[i]) Recommended Values
Rudder behind stern (skek):	0	1.3 - 1.5 (1.5 - 2.0)
Twin-screw balance rudders:	0	2.8
Shaft brackets:	0	3.0
Skeg:	0	1.5 - 2.0
Strut (hull) bossings:	0	3.0 (2.0)
Exposed shafts:	0	2.0 - 4.0
Stabilizer fins:	0	2.8
Sonar dome:	0	2.7
Bilge keels:	0	1.4
Other:	0	

NOTE: Appendage drag is proportional to $\sum((1+K_{2i})*SAPP[i])$, where $SAPP[i]$ is the wetted area of appendage i.

Inputs for Calculating the Bow Thrust Tunnel Factor:

Tunnel diameter: 0 meters Tunnel drag coefficient: 0.004

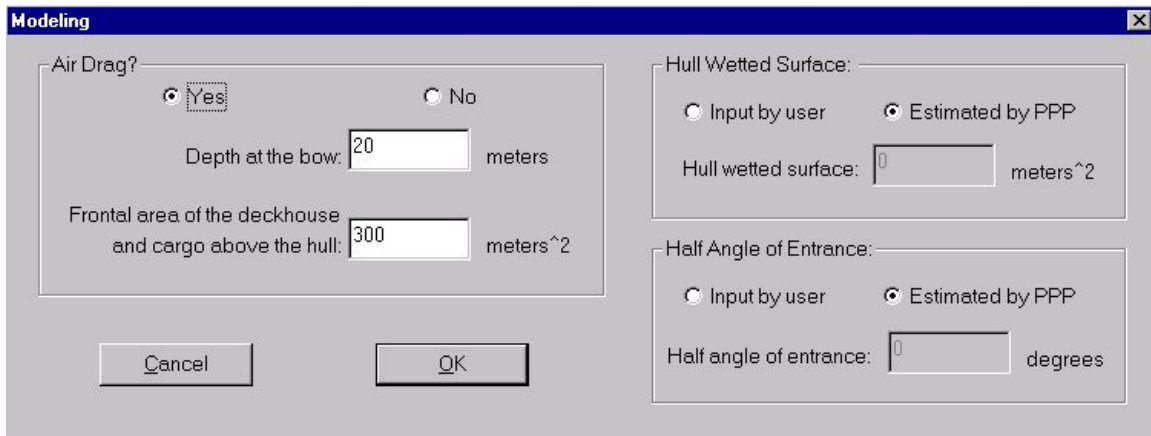
OK Cancel

$SAPP_i$ is the wetted area of appendage i in meters²

$(1 + K_{2i})$ reflects the local drag coefficient of and the local velocity at the appendage

When an option is in parentheses such as (skek) with the recommended (1.5 - 2.0) also in parenthesis, that pair is an alternative to stern and 1.3 - 1.5.

Modeling allows a choice of including an explicit air drag estimate and the use of the internal regression models for the estimation of the wetted surface and the half angle of entrance if these are not yet known in early design.



Modeling

Air Drag? ☒ Yes ☐ No

Depth at the bow: meters

Frontal area of the deckhouse and cargo above the hull: meters²

Hull Wetted Surface: ☐ Input by user ☒ Estimated by PPP

Hull wetted surface: meters²

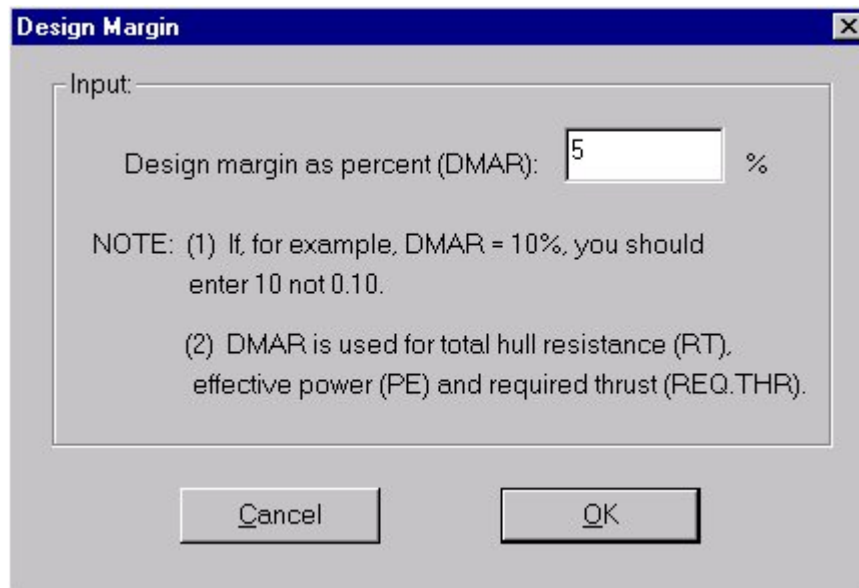
Half Angle of Entrance: ☐ Input by user ☒ Estimated by PPP

Half angle of entrance: degrees

When air drag is selected, the first two definitions apply:

- D_{BOW} depth at the bow in meters
- A_F frontal area of the deckhouse and cargo above the hull in meters²
- S hull wetted surface in meters²; or calculated by program using regression model $S(LWL, B, T_F, T_A, C_B, C_M, C_{WP}, A_{BT})$
- $\alpha_{E1/2}$ half angle of entrance of the design waterline in degrees; or calculated by program using regression model $\alpha_{E1/2}(LWL, B, T_F, T_A, C_B, C_M, C_{WP}, LCB)$

Design Margin allows the input of the design margin that is applied to the total resistance, effective power, and required propeller thrust as a percent.



Design Margin

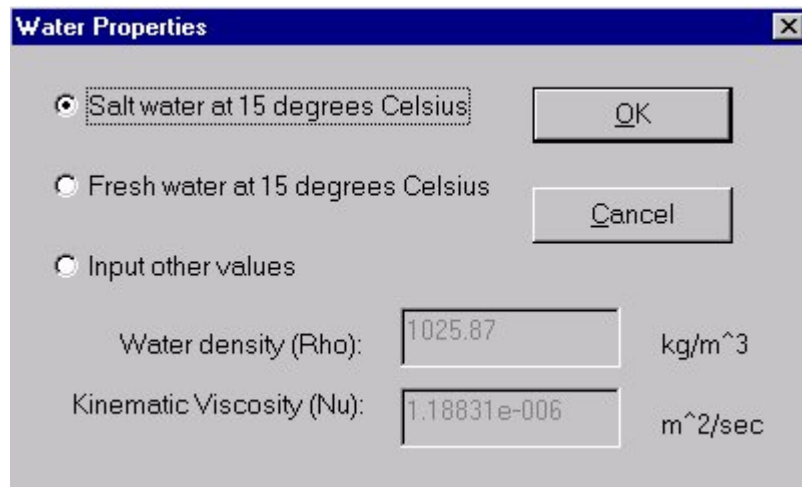
Input:

Design margin as percent (DMAR): %

NOTE: (1) If, for example, DMAR = 10%, you should enter 10 not 0.10.

(2) DMAR is used for total hull resistance (RT), effective power (PE) and required thrust (REQ. THR).

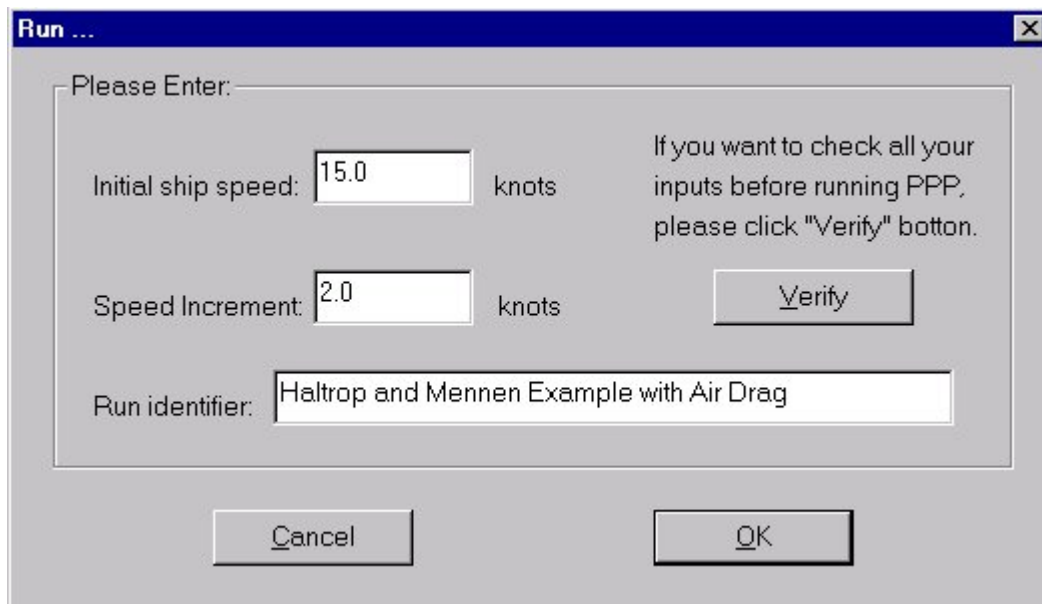
Water Properties allows the choice of fresh water @ 15C, salt water @ 15C or user-specified water properties.



The 'Water Properties' dialog box has a title bar with a close button. It contains three radio buttons: 'Salt water at 15 degrees Celsius' (selected), 'Fresh water at 15 degrees Celsius', and 'Input other values'. To the right of these are 'OK' and 'Cancel' buttons. Below the radio buttons are two input fields: 'Water density (Rho):' with the value '1025.87' and units 'kg/m^3', and 'Kinematic Viscosity (Nu):' with the value '1.18831e-006' and units 'm^2/sec'.

RUNNING PPP:

When the input process is completed, the program can be run through the **Analysis** menu or by selecting the **RunPPP** button. This will produce a final run input window where the initial speed V_o and speed increment V_i are entered in knots. The user can also enter a run specific identifier at this point.



The 'Run ...' dialog box has a title bar with a close button. It contains a 'Please Enter:' label. Below it are three input fields: 'Initial ship speed:' with the value '15.0' and units 'knots', 'Speed Increment:' with the value '2.0' and units 'knots', and 'Run identifier:' with the text 'Haltrop and Mennen Example with Air Drag'. To the right of the speed inputs is a text box with the message: 'If you want to check all your inputs before running PPP, please click "Verify" button.' Below this text box is a 'Verify' button. At the bottom of the dialog are 'Cancel' and 'OK' buttons.

OUTPUT:

A sample output report resulting from the input from the above windows follows. Notice that the added pressure resistance due to the bulb plowing the free surface R_B is negligible in this example. Also, note that the transom clears (the streamlines leave the lower edge of the transom which becomes dry) at just over 23 knots causing the added pressure resistance due to the immersed transom R_{TR} to become zero. The Required Thrust and wake fraction can be used as the propeller design requirements.

Project Name: Powering Evaluation Design

Thu Jul 09 12:39:03 1998

University of Michigan
Department of Naval Architecture and Marine Engineering

Power Prediction Program (PPP-1.8) by M. G. Parsons

Source: 1. Holtrop, J., & Mennen, G.G.J., "An Approximate Power Prediction Method," International Shipbuilding Progress, Vol.29, No.335, July, 1982.
2. Holtrop, J., "A Statistical Reanalysis of Resistance and Propulsion Data," International Shipbuilding Progress, Vol.31, No.363, Nov., 1984.

Run Identification: Holtrop and Mennen Example with Air Drag

Input Verification:

Length of Waterline LWL (m)	=	205.00
Maximum Beam on LWL (m)	=	32.00
Depth at the Bow (m)	=	20.00
Mean Draft (m)	=	10.00
Draft Forward (m)	=	10.00
Draft Aft (m)	=	10.00
Block Coefficient on LWL CB	=	0.5716
Prismatic Coefficient on LWL CP	=	0.5833
Midship Coefficient to LWL CM=CX	=	0.9800
Waterplane Coefficient on LWL CWP	=	0.7500
Center of Buoyancy LCB (% LWL; + Fwd)	=	-0.7500
Center of Buoyancy LCB (m from FP)	=	104.04
Molded Volume (m ³)	=	37497.0
Deck House/Cargo Frontal Area (m ²)	=	300.00
Water Type	=	Salt@15C
Water Density (kg/m ³)	=	1025.87
Kinematic Viscosity (m ² /s)	=	0.118831E-05
Appen. Drag (% Bare Hull Resistance)	=	5.00
Bulb Section Area at Station 0 (m ²)	=	20.00
Vertical Center of Bulb Area (m)	=	4.00
Transom Immersed Area (m ²)	=	16.00
Stern Type	=	Normally Shaped
Design Margin on RT, PE, REQ. THR (%)	=	5.00
Propulsion Type	=	SS, Conv.
Propeller Diameter (m)	=	8.00
Propeller Expanded Area Ratio Ae/Ao	=	0.7393
Wetted Surface (m ²)	=	7381.24
Half Angle of Entrance (deg)	=	12.11

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2. Holtrop, J., "A Statistical Reanalysis of Resistance and Propulsion Data," International Shipbuilding Progress, Vol.31, No.363, Nov., 1984.

Run Identification: Holtrop and Mennen Example with Air Drag

Speed, Resistance Coefficients and Frictional Resistance RF(N):

V(kts)	V(m/s)	FN	SLRATIO	CF	CR	CA	RF
15.00	7.72	0.1721	0.5784	0.001478	0.000441	0.000352	333139.8
17.00	8.75	0.1951	0.6555	0.001455	0.000472	0.000352	421443.8
19.00	9.77	0.2180	0.7326	0.001436	0.000549	0.000352	519426.3
21.00	10.80	0.2409	0.8097	0.001419	0.000674	0.000352	626970.3
23.00	11.83	0.2639	0.8869	0.001404	0.000878	0.000352	743972.3
25.00	12.86	0.2868	0.9640	0.001390	0.001108	0.000352	870339.8
27.00	13.89	0.3098	1.0411	0.001377	0.001260	0.000352	1005989.3
29.00	14.92	0.3327	1.1182	0.001366	0.001465	0.000352	1150844.6

Remaining Resistance Components (N):

V(kts)	Form RF*K1	Appendage RAPP	Wave RW	Bulb RB	Transom RTR	Correlation RA	Air Drag RAIR
15.00	53277.7	21627.6	12091.4	24.6	34017.7	79469.8	13476.4
17.00	67399.9	27907.1	36487.5	30.0	32781.2	102074.5	17309.7
19.00	83069.8	35893.0	88011.8	35.3	27316.6	127504.8	21622.2
21.00	100268.9	46238.6	180775.2	40.2	16717.9	155760.8	26413.8
23.00	118980.5	60471.7	346357.5	44.9	78.7	186842.3	31684.6
25.00	139190.0	78214.1	554702.9	49.2	0.0	220749.4	37434.5
27.00	160883.9	96303.5	759144.4	53.2	0.0	257482.1	43663.6
29.00	184050.0	119248.5	1050018.4	57.0	0.0	297040.4	50371.9

Resistance, Effective Power, Propulsion Factors and Required Thrust

V(kts)	RT(N)	PE(kW)	w	t	REQ.THR(N)	etaH	etaRR
15.00	574481.3	4433.04	0.2402	0.1834	703483.2	1.0748	0.9931
17.00	740705.4	6477.82	0.2397	0.1834	907033.5	1.0741	0.9931
19.00	948023.8	9266.33	0.2393	0.1834	1160906.0	1.0736	0.9931
21.00	1210844.9	13081.05	0.2390	0.1834	1482744.5	1.0731	0.9931
23.00	1562854.0	18491.88	0.2387	0.1834	1913798.5	1.0726	0.9931
25.00	1995714.0	25666.88	0.2384	0.1834	2443858.8	1.0722	0.9931
27.00	2439696.3	33887.09	0.2381	0.1834	2987538.8	1.0718	0.9931
29.00	2994212.3	44669.94	0.2379	0.1834	3666573.5	1.0715	0.9931

Design Margin Has Been Included in RT, PE, and REQ.THR = RT/(1-t).



PROPELLER OPTIMIZATION PROGRAM (POP1.6)

M. G. Parsons, December 1994

C.W. Culbertson, April 2005

REFERENCES:

1. Oosterveld, M. W. C., and van Oossanen, P., "Further Computer-Analyzed Data of the Wageningen B-Screw Series," *International Shipbuilding Progress*, Vol. 22, No. 251, July, 1975.
2. Parsons, M. G., "Optimization Methods for Use in Computer-Aided Ship Design," *Proceedings of the First SNAME STAR Symposium*, 1975.
3. Comstock, J. P. (Ed.), *Principles of Naval Architecture*, Vol. II, 1988 Edition, SNAME, New York, p. 182.

METHODS:

This program implements the Wageningen B-Screw Series regression equations presented by Oosterveld and van Oossanen (1). The user can either evaluate a given propeller or use mathematical programming from Parsons (2) to obtain the optimum open water efficiency propeller for a given application. Additionally, the user may specify the propeller RPM and use the mathematical programming to obtain the optimum open water efficiency propeller for a given application at this specified RPM.

In the optimization mode, the user specifies the required thrust and speed of advance. The program then uses the Nelder and Mead Simplex Search method with an External Penalty Function (2) to determine the optimum propeller. Wageningen B-Screw Series 3- through 7-bladed propellers with Reynolds number corrections are available. Optimization among blade number must be handled with separate runs for each blade number and then simple comparison. For controllable-reversible-pitch propeller estimates, the program reduces the B-Screw Series open water efficiency by 2 percent to approximate the loss of efficiency due to the larger hub.

The program includes either the Burrill five or ten percent back cavitation criterion (3). The optimization independent variables are the expanded area ratio A_e/A_o , the pitch-diameter ratio P/D_p , and the propeller diameter D_p . If the propeller RPM is free, the value needed to produce the specified thrust (within about 1 lbf.) is determined at each step of the search in an internal loop. Thus, instead of having RPM as a fourth independent variable, it is established by the equality constraint that the thrust produced must equal the required thrust. This is an implicit constraint so the required RPM is found by iteration.

In optimization with the RPM specified, the user specifies the propeller RPM in addition to the required thrust and speed of advance. Because RPM is constrained by the

user, A_e/A_o , and D_p are the only optimization independent variables and P/D_p is iterated internally to produce the specified thrust.

The inequality constraints on the variables are as follows:

$3 \leq N_{BLADE} \leq 7$	(integer, from model)
$0.3 \leq A_e/A_o \leq 1.05$	(from model)
$0.5 \leq P/D_p \leq 1.40$	(from model)
$D_{pMIN} \leq D_p \leq D_{pMAX}$	(user-specified)
$0.0 \leq J \leq 1.60$	(for ahead operation and from propeller model)
$0.0 \leq \text{thrust loading} \leq \text{Burrill 5\% or 10\% limit}$	(for ahead operation and user-selected)

The optimization is for the maximum open water efficiency propeller that will produce the required thrust at the specified speed of advance. The Burrill 5% back cavitation criterion is generally recommended.

The program does not check to see that the resulting design is "slightly to the left" of the peak of the efficiency versus J diagram as is usually recommended as a margin if the design misses the predicted operating point RPM. The user should check this on the graphical output.

In the optimization with RPM free mode, an error message will be printed if the determination of the RPM needed to produce the required thrust does not converge within 400 iterations within the computations. In the optimization with RPM specified mode, an error message will be printed if it is determined that the propeller characteristics cannot be optimized to produce the required thrust at the user specified RPM.

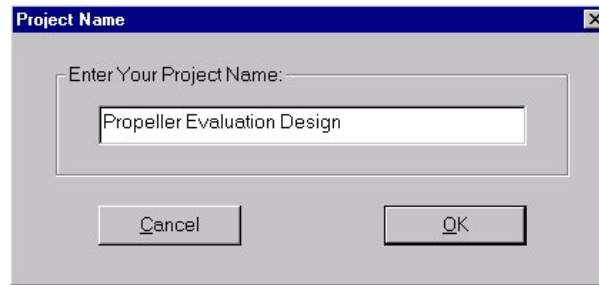
INPUT:

The input to POP is through a series of five windows within the menu **Input** as follows:

Input
Project Name
Propeller Type
Propeller Characteristics
Operating Conditions
Water Properties

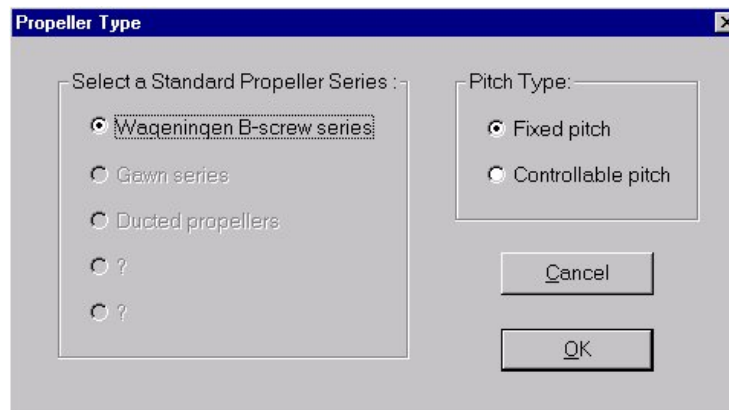
Each of these windows must be completed before an analysis can be performed. The completed input data can be saved using the **Save** button for later re-use.

Project Name provides a location for the general identification of the project being analyzed.



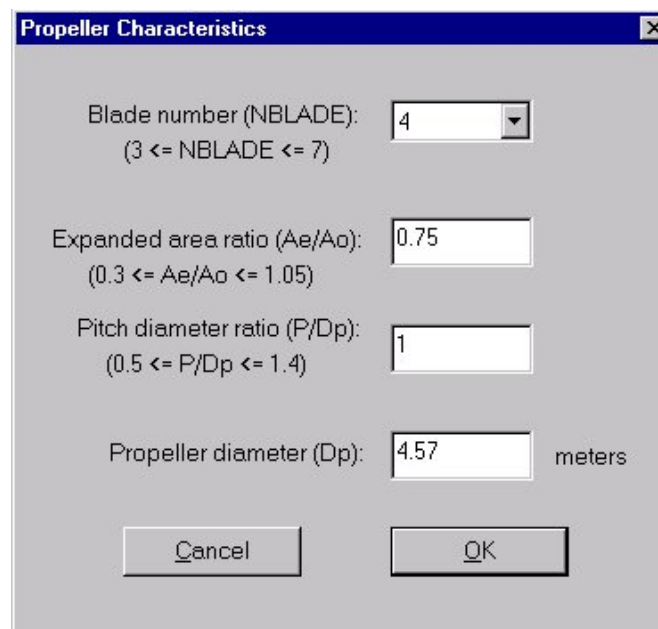
A dialog box titled "Project Name" with a close button (X) in the top right corner. It contains a label "Enter Your Project Name:" followed by a text input field containing the text "Propeller Evaluation Design". At the bottom, there are two buttons: "Cancel" and "OK".

Propeller Type allows the selection of a fixed-pitch or controllable-pitch propeller. At this time, only the Wageningen B-Screw Series is supported. As noted above, the open water efficiency is reduced by 2 percent for the controllable-pitch propeller option to approximate the effect of the larger hub.



A dialog box titled "Propeller Type" with a close button (X) in the top right corner. It is divided into two main sections. The left section, labeled "Select a Standard Propeller Series:", contains a list of radio button options: "Wageningen B-screw series" (which is selected), "Gawn series", "Ducted propellers", and two options marked with "?". The right section, labeled "Pitch Type:", contains two radio button options: "Fixed pitch" (which is selected) and "Controllable pitch". At the bottom, there are two buttons: "Cancel" and "OK".

Propeller Characteristics obtains the blade number and the basic parameters for the propeller. If the optimization option is being used, the parameters are the initial values used by the search algorithm. If the evaluation option is being used, the parameters are those evaluated by the program.



A dialog box titled "Propeller Characteristics" with a close button (X) in the top right corner. It contains four input fields with labels and ranges: "Blade number (NBLADE):" with a dropdown menu showing "4" and a range of "(3 <= NBLADE <= 7)"; "Expanded area ratio (A_e/A_o):" with a text input field showing "0.75" and a range of "(0.3 <= A_e/A_o <= 1.05)"; "Pitch diameter ratio (P/Dp):" with a text input field showing "1" and a range of "(0.5 <= P/Dp <= 1.4)"; and "Propeller diameter (Dp):" with a text input field showing "4.57" followed by the unit "meters". At the bottom, there are two buttons: "Cancel" and "OK".

N_{BLADE}	number of blades on propeller, 3 through 7 supported
A_e/A_o	initial point for search or evaluated expanded area ratio
P/D_p	initial point for search or evaluated pitch diameter ratio
D_p	initial point for search or evaluated propeller diameter in meters

Operating Conditions obtains the ship speed and wake fraction. It also allows the selection of the 5% or 10% Burrill back cavitation constraint and obtains the submergence of the shaft centerline needed to calculate the Cavitation Number at the blade tip.

Operating Conditions

Ship Speed (V_k): 15 knots

Wake fraction (w): .21067

Burrill percent of back cavitation:
☒ 5% ☐ 10%

Depth of shaft centerline below waterline: 4.39 meters

Cancel OK

V_k	ship speed in knots
w	average longitudinal wake fraction
H_{WLM}	depth of shaft centerline below waterline in meters

Water Properties allows the choice of fresh water @ 15C, salt water @ 15C or user-specified water properties.

Water Properties

☒ Salt water at 15 degrees Celsius

☐ Fresh water at 15 degrees Celsius

☐ Input other values

Water density (ρ): 1025.87 kg/m³

Kinematic viscosity (ν): 1.18831e-006 m²/sec

OK Cancel

RUNNING POP:

When the input process is completed, the program can be run through the **Analysis** menu or by selecting the **RunPOP** button. This will produce a final run control window where the selection is made for design optimization with RPM free, the evaluation of the specified operating condition, or design optimization with the RPM specified by the user. When either optimization mode is selected, the thrust required from the propeller and a minimum and maximum propeller diameter must be specified. When the optimization with RPM specified mode is selected, the RPM must also be specified. The thrust required is also needed in the evaluation mode. The user can enter a run specific identifier at this point for the linking and identification of all output.

D_{pMIN}	Minimum propeller diameter in meters. Typically this is about 25% of D_{pMAX} for fixed-pitch propellers and about 35% of D_{pMAX} for controllable-pitch propellers.
D_{pMAX}	Maximum propeller diameter permitted by hull and required clearances, in meters.
RPM	RPM of the propeller, if specified by the user. If the specified RPM cannot achieve the required thrust, an error message is printed.
Treq	Required thrust per propeller in kiloNewtons; $R_T/(1 - t)$

Run POP ...

Options:

- ☒ Optimal Design
- ☐ Evaluation
- ☐ Optimal Design at RPM Specified

"Optimal Design" requires user specified Thrust, Dpmin and Dpmax.

"Evaluation" requires user specified RPM and Propeller Characteristics.

"Optimal Design at RPM Specified" requires Thrust, RPM, Dpmin and Dpmax.

Note:

If you want to check all your inputs before running POP, please click "Verify" button.

Verify

RPM Specification:

0 RPM

Required Thrust

444.822 kN

Constraints on propeller diameter:

Propeller minimum diameter (Dpmin): 2 meters

Propeller maximum diameter (Dpmax): 4.6 meters

Run identifier: Test Optimization

Cancel OK

When the analysis process is completed, the program can be used to produce an open water efficiency η_0 , K_T , and K_Q versus J plot through the **Plot** option in the **Analysis** menu or by selecting the **Plot K_T , K_Q , η_0 versus J** button.

OUTPUT:

Sample output reports and plots resulting from the three options are shown next. Note that the operating point is indicated on each plot by a vertical line drawn at the operating advance coefficient J .

If the selected cavitation constraint is violated at the condition requested in the evaluation mode, POP prints a warning on the output.

University of Michigan
Department of Naval Architecture and Marine Engineering

Propeller Optimization Program (POP-1.6) by M.G. Parsons & C.W. Culbertson

- Source: 1. Oosterveld, M. W. C., and Van Oossanen, P.,
"Further Computer-Analyzed Data of the Wageningen
B-Screw Series", International Shipbuilding
Progress, Vol. 22, No. 251, July, 1975.
2. Parsons, M. G., "Optimization Methods for Use in
Computer-Aided Ship Design", Proceedings of the
First SNAME STAR Symposium, 1975

Wageningen B-Screw Series Propeller Preliminary Design

*** Eta 0 Reduced by 2% When Controllable Pitch ***

Run Identification: Test of Optimization with RPM Free

Input Data:

Optimization Run

Fixed-Pitch Propeller	
Number of Propeller Blades	= 5
Required Propeller Thrust (kN)	= 900.0
Ship Speed V_k (knots)	= 20.00
Wake Fraction w	= 0.185
Depth of Shaft below Waterline (m)	= 4.00
Water Type	= salt@15C
Water Density ρ (kg/m ³)	= 1025.87
Kinematic Viscosity ν (m ² /sec)	= 0.118831E-05
Burrill Back Cavitation Constraint	= 5%
Initial Expanded Area Ratio A_e/A_o	= 0.750
Initial Pitch Diameter Ratio P/D_p	= 0.900
Initial Propeller Diameter D_p (m)	= 6.00
Minimum Diameter Constraint D_{pmin} (m)	= 1.50
Maximum Diameter Constraint D_{pmax} (m)	= 6.00

Optimal Design Results:

Propeller Diameter D_p (m)	= 6.00
Propeller Pitch P (m)	= 5.42
Expanded Area Ratio A_e/A_o	= 0.7802
Pitch Diameter Ratio P/D_p	= 0.9039
Propeller Revolutions per Minute (rpm)	= 125.81
Advance Coefficient J	= 0.6673
Thrust Coefficient K_T	= 0.1544
Torque Coefficient K_Q	= 0.02365
Propeller Open Water Efficiency η_0	= 0.693
Propeller Thrust (kN)	= 900.0
Reynolds Number R_N	= 0.538E+08
Cavitation Number σ	= 0.3291
Optimization Search Evaluation Count	= 70

University of Michigan
Department of Naval Architecture and Marine Engineering
Propeller Optimization Program (POP-1.6) by M.G. Parsons & C.W. Culbertson

Wageningen B-Screw Series Propeller Characteristics

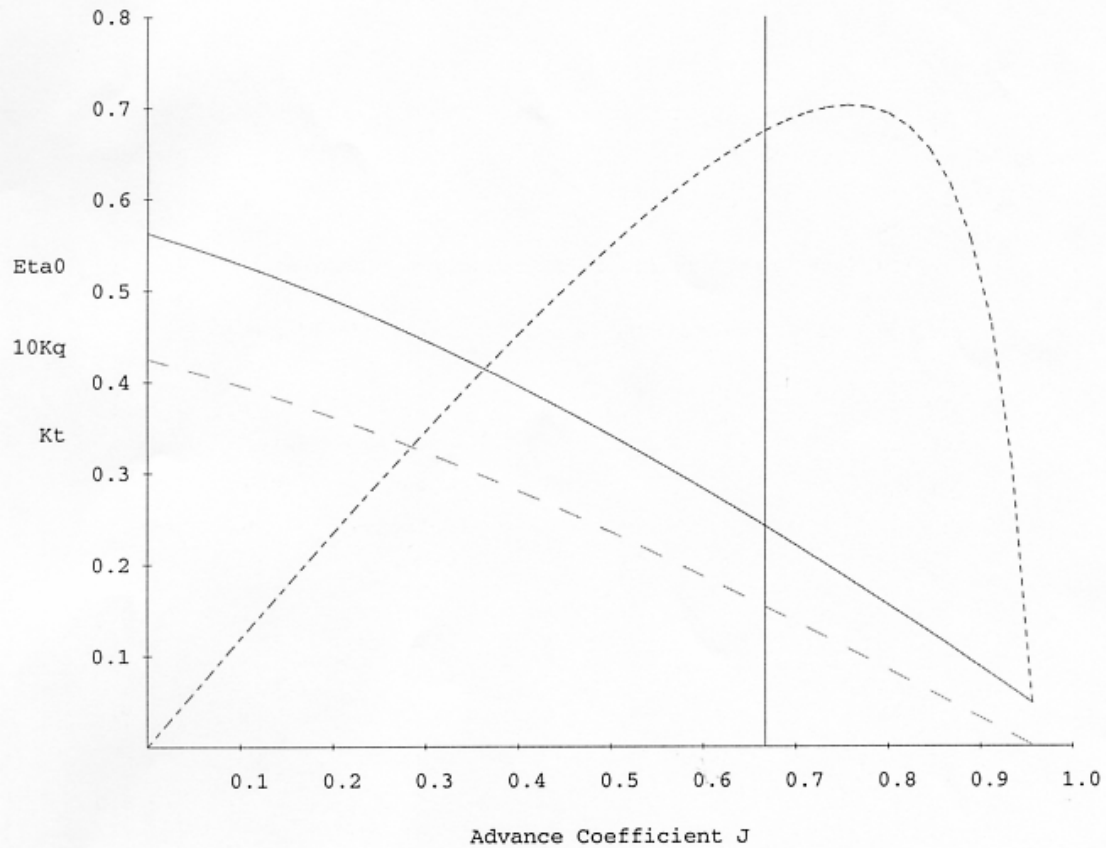
Run Identification: Test of Optimization with RPM Free

 D_p (m) = 6.00 A_e/A_o = 0.780 P/D_p = 0.904 J = 0.667

——— 10Kq

- - - - - Kt

- - - - - Eta0



University of Michigan
Department of Naval Architecture and Marine Engineering

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Source: 1. Oosterveld, M. W. C., and Van Oossanen, P.,
"Further Computer-Analyzed Data of the Wageningen
B-Screw Series", International Shipbuilding
Progress, Vol. 22, No. 251, July, 1975.
2. Parsons, M. G., "Optimization Methods for Use in
Computer-Aided Ship Design", Proceedings of the
First SNAME STAR Symposium, 1975

Wageningen B-Screw Series Propeller Preliminary Design

*** Eta 0 Reduced by 2% When Controllable Pitch ***

Run Identification: Test of Optimization with RPM Specified

Input Data:

Optimization Run with RPM Set

Fixed-Pitch Propeller	
Number of Propeller Blades	= 5
Required Propeller Thrust (kN)	= 900.0
Ship Speed V_k (knots)	= 20.00
Wake Fraction w	= 0.185
Depth of Shaft below Waterline (m)	= 4.00
Water Type	= salt@15C
Water Density ρ (kg/m^3)	= 1025.87
Kinematic Viscosity ν (m^2/sec)	= 0.118831E-05
Burrill Back Cavitation Constraint	= 5%
Initial Expanded Area Ratio A_e/A_o	= 0.750
Initial Pitch Diameter Ratio P/D_p	= 0.900
Initial Propeller Diameter D_p (m)	= 6.00
Minimum Diameter Constraint D_{pmin} (m)	= 1.50
Maximum Diameter Constraint D_{pmax} (m)	= 6.00
RPM Specified (rpm)	= 136.00

Optimal Design Results:

Propeller Diameter D_p (m)	= 5.90
Propeller Pitch P (m)	= 4.98
Expanded Area Ratio A_e/A_o	= 0.7908
Pitch Diameter Ratio P/D_p	= 0.8427
Propeller Revolutions per Minute (rpm)	= 136.00
Advance Coefficient J	= 0.6270
Thrust Coefficient K_T	= 0.1406
Torque Coefficient K_Q	= 0.02051
Propeller Open Water Efficiency η_0	= 0.684
Propeller Thrust (kN)	= 900.0
Reynolds Number R_N	= 0.569E+08
Cavitation Number σ	= 0.2934
Optimization Search Evaluation Count	= 24

University of Michigan
Department of Naval Architecture and Marine Engineering
Propeller Optimization Program (POP-1.6) by M.G. Parsons & C.W. Culbertson

Wageningen B-Screw Series Propeller Characteristics

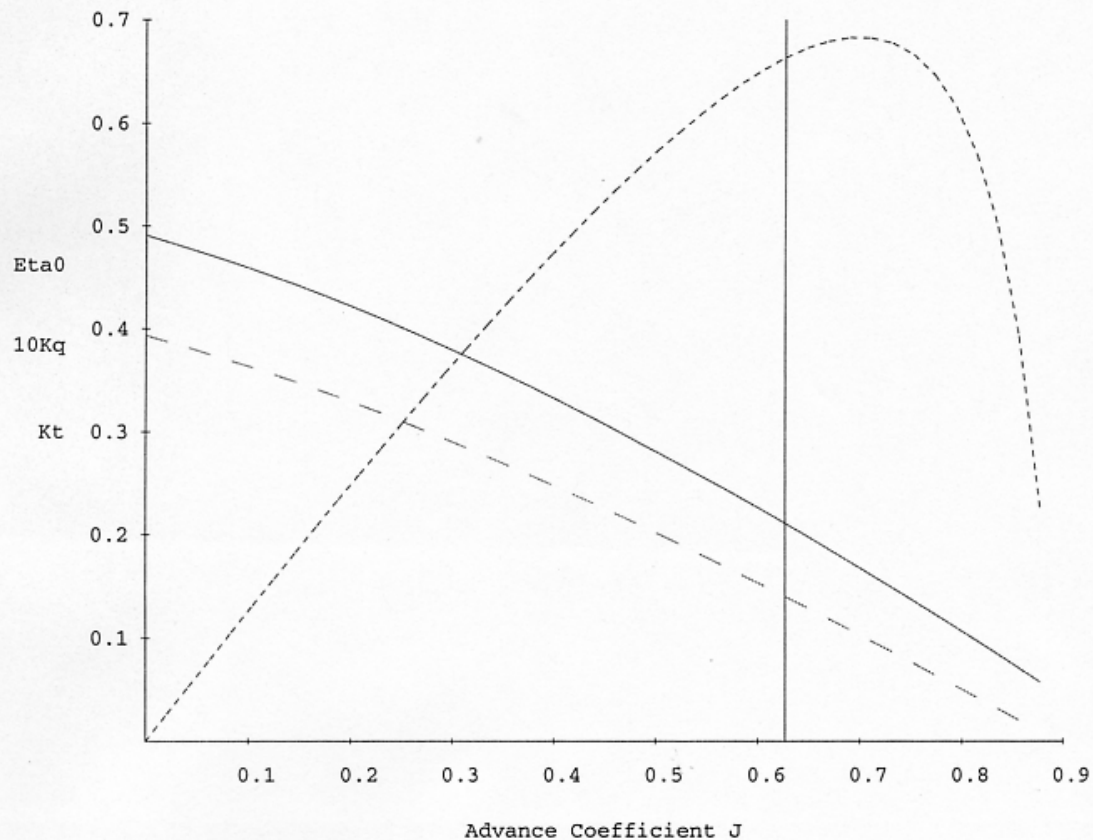
Run Identification: Test of Optimization with RPM Specified

 D_p (m) = 5.90 A_e/A_o = 0.791 P/D_p = 0.843 J = 0.627

————— 10Kq

- - - - - Kt

- - - - - Eta0



University of Michigan
Department of Naval Architecture and Marine Engineering

Propeller Optimization Program (POP-1.6) by M.G. Parsons & C.W. Culbertson

Source: 1. Oosterveld, M. W. C., and Van Oossanen, P.,
"Further Computer-Analyzed Data of the Wageningen
B-Screw Series", International Shipbuilding
Progress, Vol. 22, No. 251, July, 1975.
2. Parsons, M. G., "Optimization Methods for Use in
Computer-Aided Ship Design", Proceedings of the
First SNAME STAR Symposium, 1975

Wageningen B-Screw Series Propeller Preliminary Design

*** Eta 0 Reduced by 2% When Controllable Pitch ***

Run Identification: Test of Evaluation Mode

Input Data:

Evaluation Run

Fixed-Pitch Propeller	
Number of Propeller Blades	= 5
RPM Specified (rpm)	= 126.00
Ship Speed V_k (knots)	= 20.00
Wake Fraction w	= 0.185
Depth of Shaft below Waterline (m)	= 4.00
Water Type	= salt@15C
Water Density ρ (kg/m ³)	= 1025.87
Kinematic Viscosity ν (m ² /sec)	= 0.118831E-05
Burrill Back Cavitation Constraint	= 5%
Expanded Area Ratio A_e/A_o	= 0.750
Pitch Diameter Ratio P/D_p	= 0.900
Propeller Diameter D_p (m)	= 6.00

Output Results:

Propeller Diameter D_p (m)	= 6.00
Propeller Pitch P (m)	= 5.40
Expanded Area Ratio A_e/A_o	= 0.7500
Pitch Diameter Ratio P/D_p	= 0.9000
Propeller Revolutions per Minute (rpm)	= 126.00
Advance Coefficient J	= 0.6660
Thrust Coefficient K_T	= 0.1536
Torque Coefficient K_Q	= 0.02353
Propeller Open Water Efficiency η_o	= 0.692
Propeller Thrust (kN)	= 900.1
Reynolds Number R_N	= 0.518E+08
Cavitation Number σ	= 0.3279

University of Michigan
Department of Naval Architecture and Marine Engineering
Propeller Optimization Program (POP-1.5) by M. G. Parsons

Wageningen B-Screw Series Propeller Characteristics

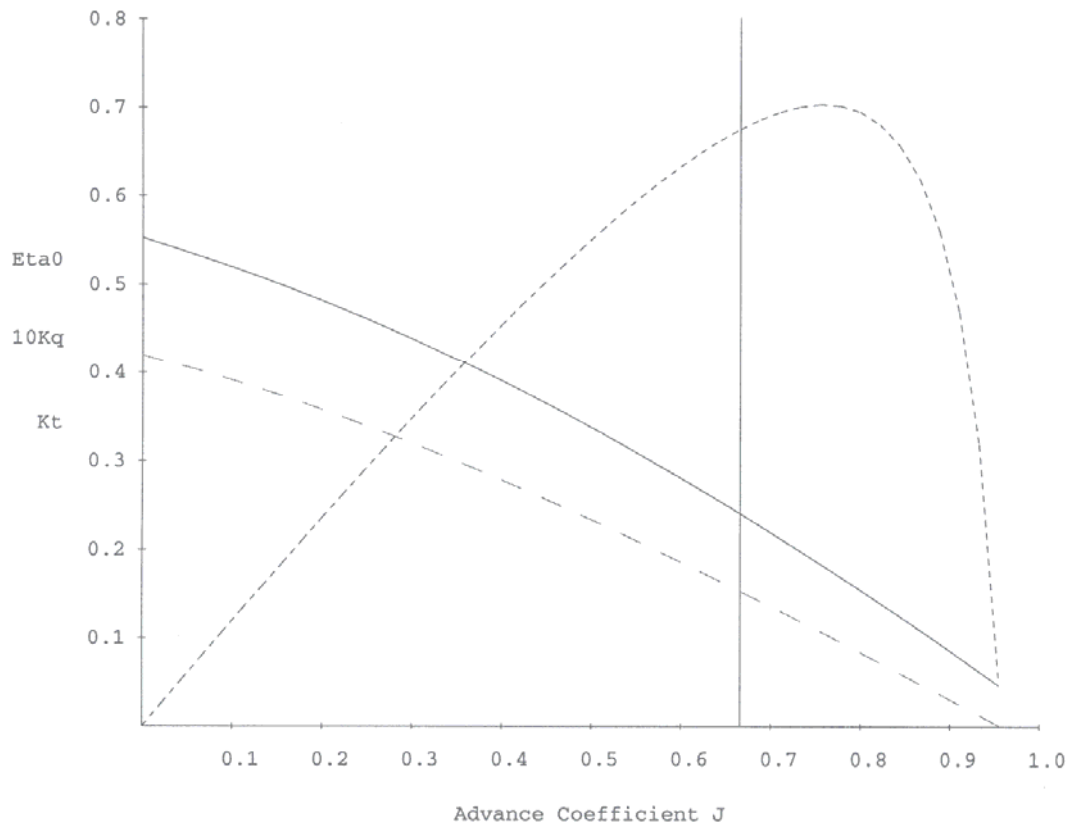
Run Identification: Test of Evaluation Mode

 D_p (m) = 6.00 A_e/A_o = 0.750 P/D_p = 0.900 J = 0.666

————— 10Kq

————— Kt

----- Eta0





MANEUVERING PREDICTION PROGRAM (MPP1.3)

M. G. Parsons, December, 1994

REFERENCES:

1. Clarke, D., Gedling, P., and Hine, G., "The Application of Manoeuvring Criteria in Hull Design Using Linear Theory," *Transactions RINA*, Vol. 124, 1982.
2. Lyster, C. A., and Knights, H. L., "Prediction Equations for Ships' Turning Circles," *Transactions NECIES* 1978-1979.
3. Parsons, M. G., "Informal Notes for NA530 Automatic Control in Naval Architecture and Marine Engineering," Vol. I, "Classical Control," and Vol. II "Modern Control"
4. Fugino, M., "Maneuverability in Restricted Waters: State of the Art," The University of Michigan NAME Report No. 184, Aug., 1976.
5. Inoue, S., Hirano, M., and Kijima, K., "Hydrodynamic Derivatives on Ship Manoeuvring," *International Shipbuilding Progress*, Vol. 28, No. 321, 1981.

METHODS:

This program offers two main options: **Linear Evaluation** which implements the methods proposed by Clarke, Gedling, and Hine (1) for the assessment of course stability, turning ability, and controllability by a helmsman; and **Turning Prediction** which implements the multiple linear regression equations presented by Lyster and Knights (2) for the estimation of turning circle characteristics. The user should consult the Clarke paper prior to using the program. If the **Linear Evaluation** option is used, the user can also obtain the linearized maneuvering equations of motion in state variable form (3).

In the **Linear Evaluation** option, the linear derivatives are estimated using the multiple linear regression equations presented by Clarke et al (1). The velocity derivatives were obtained from 72 sets of data from rotating arm and planar motion tests. Acceleration derivatives were obtained from planar motion test data. The independent variables include T/LWL , $C_B \cdot B/T$, B/LWL , and B/T . Clarke feels that these results are at least as effective as the semi-empirical equations developed by Wagner Smitt, Norrbin, or Inoue. Water depth corrections were added by regressing the data presented by Fugino (4) for the *Tokyo Maru* ($C_B = 0.805$) and the *Mariner* ($C_B = 0.589$). Independent variables in these analyses were C_B and inverse powers of $F = (H/T) - 1$. These can be used for water depths down to $H/T = 1.3$ and should be considered unreliable below this value. The trim corrections presented by Inoue (5) are used for the velocity derivatives. The independent variable in these corrections is the trim to mean draft ratio. The various time constants and gains in the first- and second-order Nomoto's equations are calculated from the linear derivatives.

The quantities calculated to allow evaluation of the maneuvering performance of the vessels are as follows:

- Inverse time constant ($1/T'$) and inverse gain ($1/K'$). Turning ability increases with decreasing $1/K'$ while course stability and quickness of response to helm increases with $1/T'$.
- Clarke's turning index P . This quantity is derived from the linear equations as the amount of turn achieved in one ship length using a unit step rudder. A minimum value of 0.3 (equivalent to about 10 degrees for a 35 degree step rudder, within the limits of the linearized equations of motion) is recommended with values as low as 0.2 considered acceptable for large tankers.
- Bare hull dynamic stability criterion C . This must be positive for open-loop, controls-fixed straight-line stability of the bare hull.
- Phase margin applicable to the closed-loop steering system. The phase margin of the ship transfer function, including the steering gear, is calculated. Nomoto notes that by anticipating the ship's turning, a typical helmsman can apply between 20 and 30 degrees of phase advance. Thus, the ship with steering gear cannot have more than this amount of negative phase margin for the combination of the ship and helmsman to be closed-loop stable on course. Thus, a phase margin greater than about -20 degrees is recommended for reasonable human control.

All output is nondimensional following Clarke (1) except the phase margin which is expressed in degrees.

In the completely independent **Turning Prediction** option, the regression equations developed by Lyster and Knights (2) from sea trial results are used to estimate turning characteristics for a user-input rudder angle and approach speed. The dependence of the turns of single screw ships on the direction of propeller rotation and the ratio of the mean draft to the design draft were not implemented. The option produces estimates of the following:

steady turning diameter (meters)
tactical diameter (meters)
advance (meters)
transfer (meters)
steady speed in the turn (knots)

This model is for deep water and includes no dependence on water depth. The range of the ship parameters included in the Lyster and Knights' regression data set is as follows:

single screw			twin screw		
54.9 ≤	LBP	≤ 329.2 m	76.2 ≤	LBP	≤ 225.6 m
0.56 ≤	C _B	≤ 0.87	0.42 ≤	C _B	≤ 0.62
10.0 ≤	rudder angle	≤ 45 degrees	10.0 ≤	rudder angle	≤ 35 degrees
5.56 ≤	L/B	≤ 9.09	5.00 ≤	L/B	≤ 16.67
0.00 ≤	trim/LBP	≤ 0.05	-0.01 ≤	trim/LBP	≤ 0.03
0.01 ≤	A _R	≤ 0.04	0.01 ≤	A _R	≤ 0.02 per rudder
-0.11 ≤	A _B	≤ 0.04	-0.10 ≤	A _B	≤ 0.0
0.060 ≤	F _n	≤ 0.298	0.074 ≤	F _n	≤ 0.655

This model should be used with CAUTION outside this range.

INPUT:

The input to MPP is through a series of five windows within the menu **Input** as follows:

Input
Project Name
Vessel Characteristics
Steering Characteristics
Operating Conditions
Water Properties

Each of these windows must be completed before an analysis can be performed. The completed input data can be saved using the **Save** button for later re-use.

Project Name provides a location for the general identification of the project being analyzed.

The screenshot shows a standard Windows-style dialog box titled "Project Name". Inside the dialog, there is a label "Enter Your Project Name:" and a text box containing the text "Maneuvering Evaluation Design". At the bottom of the dialog, there are two buttons: "Cancel" and "OK".

Vessel Characteristics obtains the basic dimensions, form coefficients, LCG, yaw radius of gyration, and bow profile information for the hull.

Vessel Characteristics

Basic Geometric Features:

Length on waterline (LWL): 148 meters

Maximum beam (B): 32.3 meters

Draft forward (TF): 9 meters

Draft aft (TA): 9 meters

Block coefficient on LWL (CB): 0.7

Others:

Longitudinal center of gravity (LCG): 0 %
(in %LWL from midships; + forward)

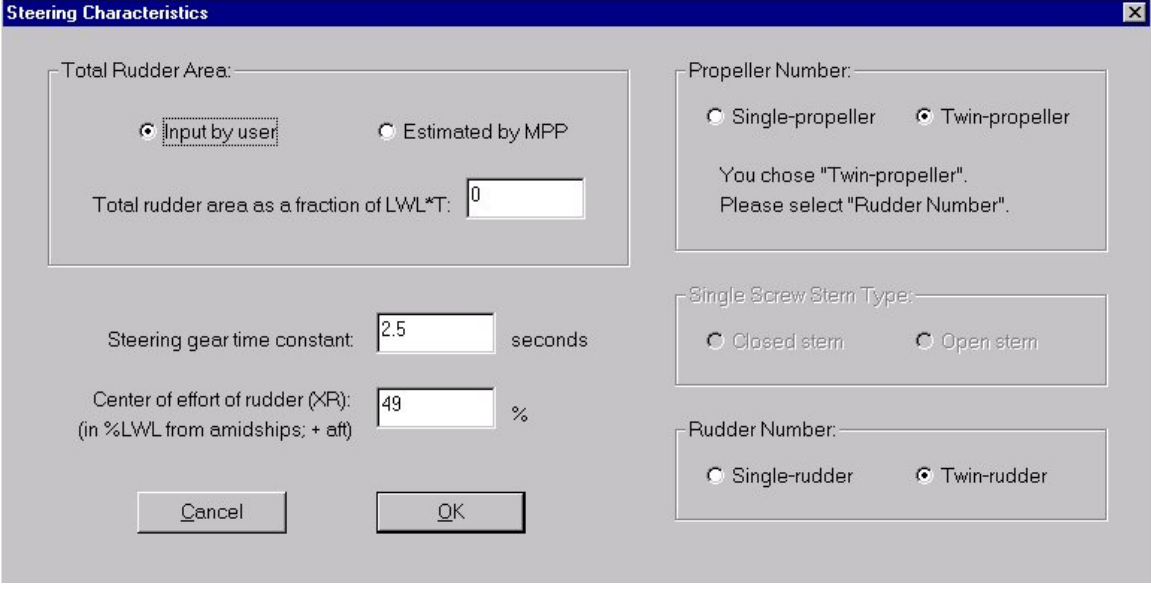
Yaw radius of gyration K33: 0.25
(as a fraction of LWL)

Non-dimensional difference of bow profile from a plumb bow: 0.016
(as a fraction of LWL*T)

Cancel OK

LWL	length on waterline in meters
B	maximum beam on waterline in meters
T _F	draft forward in meters
T _A	draft aft in meters
C _B	block coefficient to LWL
LCG	longitudinal center of gravity in %LWL from amidships; + forward
GYRAD	yaw radius of gyration k ₃₃ as a fraction of LWL; default value 0.25
A _B	difference between the submerged bow profile area on the centerplane and that of a plumb bow as fraction of LWL*T; i.e., $A_B = \text{area projects cutaway area aft of with a raked stem and/or cutaway forefoot}$

Steering Characteristics obtains the rudder area, steering gear time constant, position of the rudder center of effort (typically the 1/4 chord point from the leading edge; the rudder post), and selection of single or twin screw. The rudder area can be input either as a fraction of LWL*T or generated automatically by PPP as the minimum rudder area specified by Det Norske Veritas for commercial vessels. For single screw, the stern type must be designated as a closed or open stern. For twin screw, the number of rudders must be designated as one or two.



Steering Characteristics

Total Rudder Area:

☒ Input by user ☐ Estimated by MPP

Total rudder area as a fraction of LWL*T:

Steering gear time constant: seconds

Center of effort of rudder (X_R): %
(in %LWL from amidships; + aft)

Propeller Number:

☐ Single-propeller ☒ Twin-propeller

You chose "Twin-propeller".
Please select "Rudder Number".

Single Screw Stem Type:

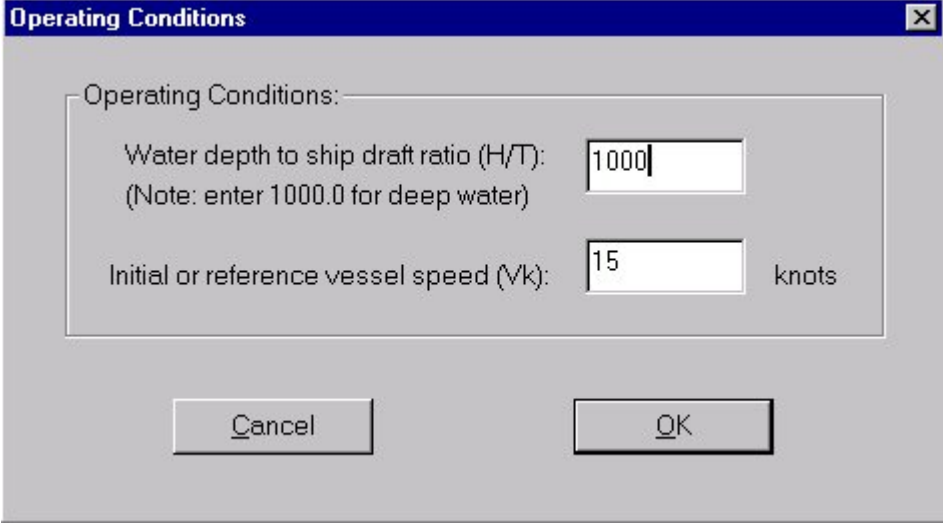
☐ Closed stem ☐ Open stem

Rudder Number:

☐ Single-rudder ☒ Twin-rudder

A_R total rudder area as fraction of LWL*T; i.e., $A_R = A/(LWL*T)$; option
 is calculated from the Det Norske Veritas **minimum** rudder area
 T_E steering gear time constant in seconds; default value is 2.5 s
 X_R center of effort of rudder in %LWL from amidships; + aft;
 i.e., about 49.0

Operating Conditions obtains the water depth and the initial vessel speed.



Operating Conditions

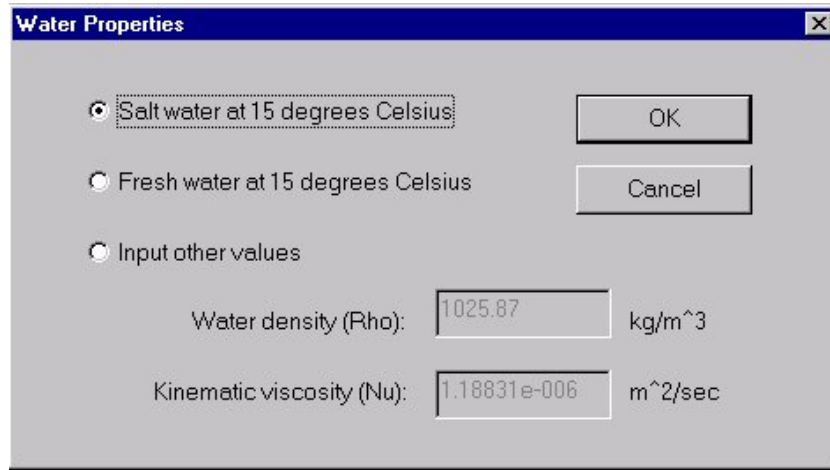
Operating Conditions:

Water depth to ship draft ratio (H/T):
(Note: enter 1000.0 for deep water)

Initial or reference vessel speed (V_k): knots

H/T water depth to ship draft ratio (H/T); input 1000 for deep water
 V_K initial or reference ship speed in knots

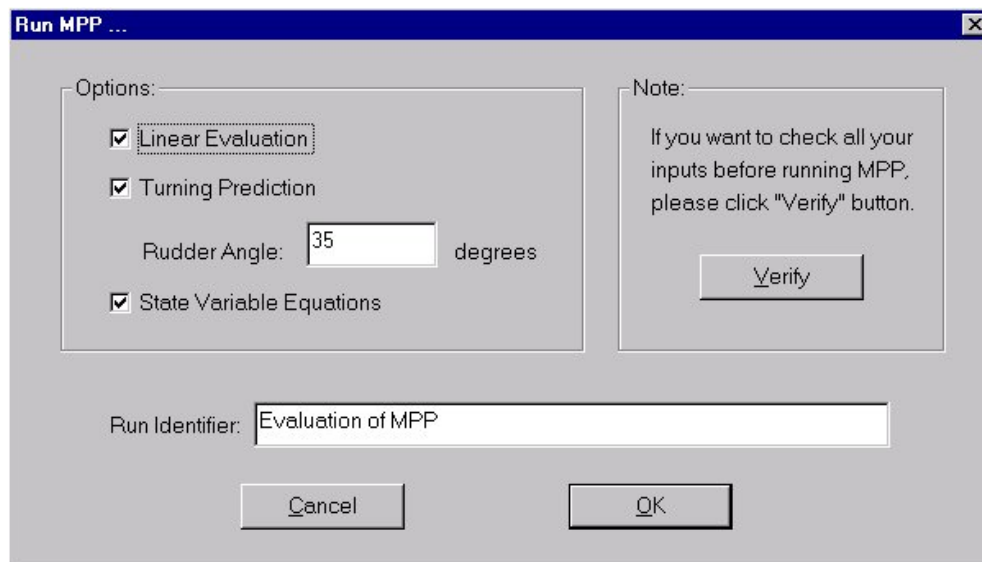
Water Properties allows the choice of fresh water @ 15C, salt water @ 15C or user-specified water properties.



The **Water Properties** dialog box has a title bar with a close button. It contains three radio buttons:
☒ Salt water at 15 degrees Celsius
☐ Fresh water at 15 degrees Celsius
☐ Input other values
 To the right of these are **OK** and **Cancel** buttons.
 Below the radio buttons, there are two input fields:
 Water density (Rho): 1025.87 kg/m³
 Kinematic viscosity (Nu): 1.18831e-006 m²/sec

RUNNING MPP:

When the input process is completed, the program can be run through the **Analysis** menu or by selecting the **RunMPP** button. This will produce a final run input window where the selection is made for the **Linear Evaluation**, **Turning Prediction**, and **State Variable Equations** options. With selection of the **Turning Prediction** option, a positive rudder angle must be specified. The user can also enter a run specific identifier at this point.



The **Run MPP ...** dialog box has a title bar with a close button. It is divided into two main sections.
 On the left, under the heading "Options:", there are three checked checkboxes:
☒ Linear Evaluation
☒ Turning Prediction
☒ State Variable Equations
 Below these is a text input field for "Rudder Angle:" with the value "35" and the unit "degrees".
 On the right, under the heading "Note:", there is a text box containing the message: "If you want to check all your inputs before running MPP, please click 'Verify' button." Below this text is a **Verify** button.
 At the bottom of the dialog, there is a text input field for "Run Identifier:" with the text "Evaluation of MPP".
 At the very bottom are **Cancel** and **OK** buttons.

OUTPUT:

A sample output report resulting from the input from the above windows follows.

University of Michigan
Department of Naval Architecture and Marine Engineering

Maneuvering Prediction Program (MPP-1.3) by M.G. Parsons

References: Clarke, D., Gedling, P., and Hine, G.,
"The Application of Manoeuvring Criteria in Hull
Design using Linear Theory," Trans. RINA, 1983
Lyster, C., and Knights, H. L.,
"Prediction Equations for Ships' Turning Circles,"
Trans. NECIES, 1978-1979

Run Identification: Evaluation of MPP

Input Verification:

Length of Waterline LWL (m)	=	148.00
Maximum Beam on LWL (m)	=	32.30
Mean Draft (m)	=	9.00
Draft Forward (m)	=	9.00
Draft Aft (m)	=	9.00
Block Coefficient on LWL CB	=	0.7000
Molded Volume (m ³)	=	30116.52
Center of Gravity LCG (%LWL; + Fwd)	=	0.0000
Center of Gravity LCG (m from FP)	=	74.00
Midships to Rudder CE XR (%LWL; + Aft)	=	49.0000
Rudder Center of Effort XR (m from FP)	=	146.52
Initial Ship Speed (knots)	=	15.00
Initial Ship Speed (m/s)	=	7.7166
Water Type	=	Salt@15C
Water Density (kg/m ³)	=	1025.87
Kinematic Viscosity (m ² /s)	=	0.118831E-05
Yaw Radius of Gyration K33/LWL	=	0.2500
Water Depth to Ship Draft Ratio H/T	=	1000.00
Steering Gear Time Constant (s)	=	2.50
Total Rudder Area - Fraction of LWL*T	=	0.0219
Number of Propellers	=	2
Number of Rudders	=	2
Submerged Bow Area - Fraction of LWL*T	=	0.0160

University of Michigan
 Department of Naval Architecture and Marine Engineering
 Maneuvering Prediction Program (MPP-1.3) by M.G. Parsons

*** Linear Maneuvering Criteria Option ***

Reference: Clarke, D., Gedling, P., and Hine, G.,
 "The Application of Manoeuvring Criteria in Hull
 Design using Linear Theory," Trans. RINA, 1983

Run Identification: Evaluation of MPP

Linear Maneuvering Derivatives

Nondimensional Mass	M prime	=	0.018580
Nondimensional Mass Moment	I sub zz	=	0.001161
Sway Velocity Derivative	Y sub v	=	-0.024491
Sway Acceleration Derivative	Y sub v dot	=	-0.013465
Yaw Velocity Derivative	N sub v	=	-0.006917
Yaw Acceleration Derivative	N sub v dot	=	-0.001080
Sway Velocity Derivative	Y sub r	=	0.004154
Sway Acceleration Derivative	Y sub r dot	=	-0.001205
Yaw Velocity Derivative	N sub r	=	-0.003398
Yaw Acceleration Derivative	N sub r dot	=	-0.000628
Sway Rudder Derivative	Y sub delta	=	0.003997
Yaw Rudder Derivative	N sub delta	=	-0.001958

Time Constants and Gains for Nomoto's Equation

Dominant Ship Time Constant	T1 prime	=	-8.1946
Ship Time Constant	T2 prime	=	0.4130
Numerator Time Constant	T3 prime	=	0.8871
Numerator Time Constant	T4 prime	=	0.2273
1st Order Eqn. Time Constant	T prime	=	-8.6686
Rudder Gain Factor	K prime	=	4.5675
Rudder Gain Factor	K sub v prime	=	-2.5273
Steering Gear Time Constant	TE prime	=	0.1303

Evaluation of Turning Ability and Stability

Inverse Time Constant	1/T prime	=	0.1154
Inverse Gain Factor	1/K prime	=	0.2189
Clarke's Turning Index	P	=	0.4633
Linear Dynamic Stability Criterion	C	=	-0.0000166

Vessel is hydrodynamically open loop course unstable

Closed Loop Phase Margin with Steering Engine = 2.4258 degrees

University of Michigan
Department of Naval Architecture and Marine Engineering
Maneuvering Prediction Program (MPP-1.3) by M.G. Parsons

*** Turning Prediction Option ***

Reference: Lyster, C., and Knights, H. L.,
"Prediction Equations for Ships" Turning Circle",
Trans. NECIES, 1978-1979

Run Identification: Evaluation of MPP

Approach Speed	=	15.00 knots
Rudder Angle	=	35.00 degrees
Steady Turning Diameter	=	489.71 meters
Tactical Diameter	=	510.02 meters
Advance	=	421.76 meters
Transfer	=	219.02 meters
Steady Speed in Turn	=	9.62 knots

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***State Variable Equations Option ***

Run Identification: Evaluation of MPP

$$\begin{aligned} \frac{d}{dt} \begin{bmatrix} \psi \\ r \\ v \end{bmatrix} &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & a_{22} & a_{23} \\ 0 & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} \psi \\ r \\ v \end{bmatrix} + \begin{bmatrix} 0 \\ b_2 \\ b_3 \end{bmatrix} \delta \end{aligned}$$

$$\begin{aligned} &+ \begin{bmatrix} 0 & 0 \\ \gamma_{21} & \gamma_{22} \\ \gamma_{31} & \gamma_{32} \end{bmatrix} \begin{bmatrix} Y_{\text{external}} \\ N_{\text{external}} \end{bmatrix} \end{aligned}$$

Open Loop Dynamics Matrix Coeff.	a22 =	-1.66593
Open Loop Dynamics Matrix Coeff.	a23 =	3.48446
Open Loop Dynamics Matrix Coeff.	a32 =	0.44714
Open Loop Dynamics Matrix Coeff.	a33 =	-0.63323

Control Distribution Matrix Coeff.	b2 =	-1.19719
Control Distribution Matrix Coeff.	b3 =	-0.16973

Ext. Force Distr. Matrix Coeff.	gamma 21 =	-19.26932
Ext. Force Distr. Matrix Coeff.	gamma 22 =	571.99927
Ext. Force Distr. Matrix Coeff.	gamma 31 =	-31.93038
Ext. Force Distr. Matrix Coeff.	gamma 32 =	21.50796



SEAKEEPING PREDICTION PROGRAM (SPP1.5)

M. G. Parsons, March, 1997

REFERENCES:

1. Raff, A. I., "Program SCORES - Ship Structural Response in Waves," Ship Structures Committee Report SSC-230, 1972.
2. Price, W. G. and Bishop, R. E. D., *Probabilistic Theory of Ship Dynamics*, Halsted Press, 1974.
3. Troesch, A. W., "Naval Architecture NA440 Ship Dynamics II," Course Notes.

METHODS:

This program implements a version of the five degree-of-freedom SCORES Program written by Raff and Kaplan while at Oceanics, Inc. in the early 1970's (1). This probabilistic analysis (2, 3) involves the use of strip theory and models long-crested seas. Each hull section is represented by a Lewis Form transformation which makes it possible to run the program when only the Section Area, Design Waterline, and drafts are known. An input option also allows the approximate development of this data from early ship parameters only. The hydrodynamic calculations utilize ideal fluid theory which seriously under predicts the roll damping, which is mostly viscous. To produce acceptable roll results, the user can input an effective roll damping ratio (zeta) to account for the viscous effects in roll.

The program calculates the motion response in heave, pitch, and roll and the vertical wave bending moment, lateral wave bending moment, and hull torsional moment in a specified case of sea spectrum, ship speed, and ship heading relative to the waves. The natural frequencies and periods in heave, pitch, and roll are calculated. The Root Mean Square (RMS), average, average of the 1/3 highest or significant, and average of the 1/10 highest values are calculated for the input wave spectrum and each of the six response spectra. A design value is also produced for each of the bending moments M_i using a long-term extreme prediction due to Ochi,

$$\text{design value } M_i = \text{RMS}_i * \text{SQRT}(2 * \ln(N/\alpha)),$$

with $N = 1000$, roughly the number of waves encountered in 3+ hour peak exposure in a severe storm; and $\alpha = 0.01$, giving an approximate probability of exceedance of one percent. The Pierson-Moskowitz spectrum, ISSC two parameter spectrum, and the JONSWAP fetch-limited spectrum are available as wave inputs.

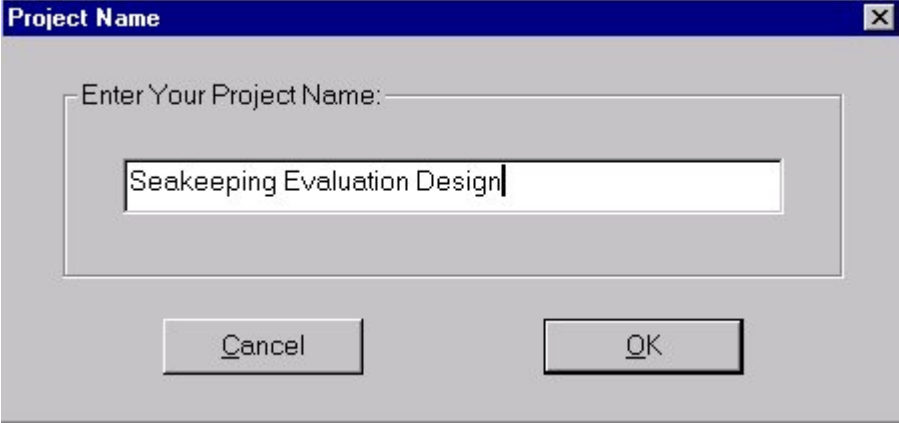
INPUT:

The input to SPP is through a series of four windows within the menu **Input** as follows:

Input
Project Name
Vessel Characteristics
Operating Conditions
Seaway Characteristics

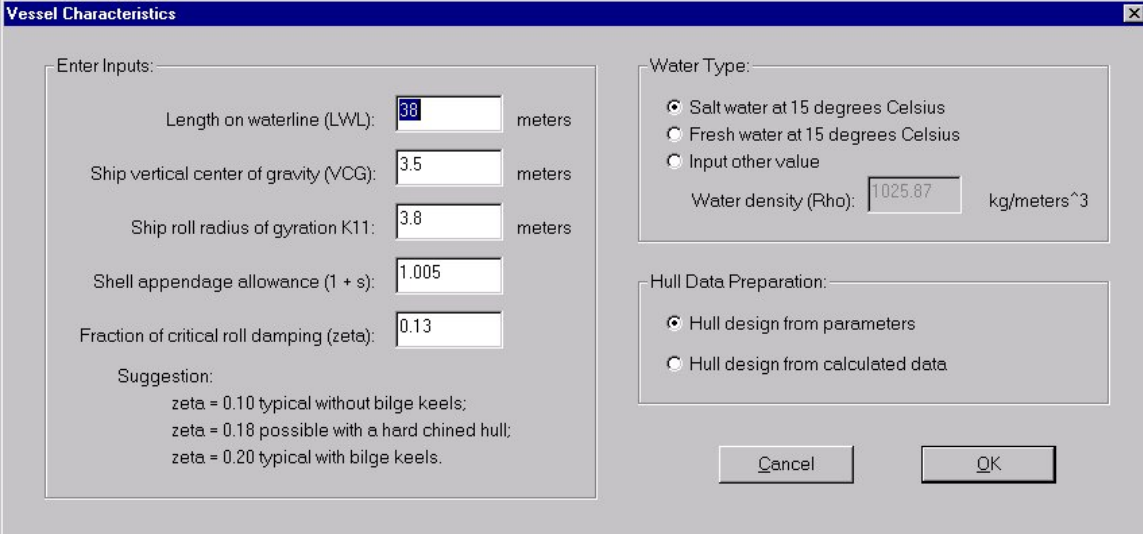
Each of these windows must be completed before an analysis can be performed. The completed input data can be saved using the **Save** button for later re-use.

Project Name provides a location for the identification of the project being analyzed.



The screenshot shows a dialog box titled "Project Name" with a close button (X) in the top right corner. Inside the dialog, there is a label "Enter Your Project Name:" followed by a text input field. The input field contains the text "Seakeeping Evaluation Design". Below the input field, there are two buttons: "Cancel" and "OK".

Vessel Characteristics obtains the basic dimensions and characteristics of the hull. It also includes the selection of the water properties from among fresh water @15C, salt water @15C, or user-specified properties.



The screenshot shows a dialog box titled "Vessel Characteristics" with a close button (X) in the top right corner. The dialog is divided into several sections. On the left, under "Enter Inputs:", there are five input fields with labels and units: "Length on waterline (LWL):" with value 38 and unit "meters"; "Ship vertical center of gravity (VCG):" with value 3.5 and unit "meters"; "Ship roll radius of gyration K11:" with value 3.8 and unit "meters"; "Shell appendage allowance (1 + s):" with value 1.005; and "Fraction of critical roll damping (zeta):" with value 0.13. Below these is a "Suggestion:" section with three lines of text: "zeta = 0.10 typical without bilge keels;", "zeta = 0.18 possible with a hard chined hull;", and "zeta = 0.20 typical with bilge keels." On the right, under "Water Type:", there are three radio buttons: "Salt water at 15 degrees Celsius" (selected), "Fresh water at 15 degrees Celsius", and "Input other value". Below the radio buttons is a text input field for "Water density (Rho):" with value 1025.87 and unit "kg/meters^3". At the bottom right, under "Hull Data Preparation:", there are two radio buttons: "Hull design from parameters" (selected) and "Hull design from calculated data". At the bottom of the dialog, there are two buttons: "Cancel" and "OK".

LWL	length on waterline in meters
VCG	ship vertical center of gravity in meters
GYRAD	ship roll radius of gyration k_{11} in meters; for typical commercial hulls this can be estimated as $k_{11} = 0.5 \cdot 0.79 \cdot B$
(1 + s)	shell appendage allowance on molded displacement; typically 1.005 for large ships
zeta	decimal fraction of critical roll damping (zeta = 0.10 is typical without bilge keels; zeta = 0.18 is possible with a hard chined hull; zeta = 0.20 is typical with bilge keels)

This input window also includes the choice of two ways to enter the hull definition for the analysis with options for **Hull design from parameters** and **Hull design from calculated data**. The first option approximates the Sectional Area Curve and Design Waterline Curve from parameters using 6th-order (degree 5) polynomials which depend upon the hull parameters. The second option is for use after the hull form has been developed and the Sectional Area Curve and Design Waterline Curve are available from hydrostatic analysis of the hull surface or offsets.

The input window for the option selected either generates or requires input of the following hull data provided at eleven stations ($I = 0$ to 10 from the FP):

BEAM(I)	station waterline full breadth (local beam) in meters
AREA(I)	station (local) submerged area in meters ²
DRAFT(I)	station draft in meters (NOTE: if the BEAM is zero, the corresponding DRAFT must also be zero.)
DWEIGHT(I)	<u>discrete</u> weight of the portion of ship centered at each station in tonnes. (NOTE: there is only half a station spacing plus any overhang, associated with stations 0 and 10.)

The station spacing is assumed to be $LWL/10$. The area and the weight must agree within 2% and the area LCB and the weight LCG must agree within 0.5% of the LWL. The input process checks these limits. The discrete weight distribution is used only to estimate the pitch mass moment of inertia and can be somewhat approximate provided it produces the correct total weight and LCG. The spreadsheet input window aids the user in developing an acceptable discrete weight distribution.

1. **Hull design from parameters** option:

In this first option, a spreadsheet appears which allows the development of an approximate Design Waterline Curve and Section Area Curve from ship parameters using 6-th order (degree 5) polynomials for each curve. This model neglects a bulbous bow, but allows a transom stern since this has significant impact on the design waterplane and, thus, seakeeping. The model is not effective in modeling hulls with significant parallel midbody. The user inputs maximum beam, C_p , C_{WP} , C_X , draft, wetted area of the transom at zero speed in m², transom beam at zero speed as a percent of maximum beam, and LCB and LCF from amidships as a percent of LWL, positive forward.

Parametric Hull Model

Parameters:

Maximum beam on LWL (B): meters

Draft (T): meters

Prismatic coefficient on LWL (Cp):

Midship coefficient (Cx = Cm):

Waterplane coefficient on LWL (Cwp):

Wetted area of transom at zero speed: meters²

Beam of transom at zero speed (%B): %

Longitudinal center of buoyancy (LCB): %
(%LWL from amidships; + fwd)

Longitudinal center of flotation (LCF): %
(%LWL from amidships; + fwd)

Hull Data Identifier:

Click "Start" button to initialize hull design on the right

Hull Design:

(1) Data in the two columns with white background can be changed. Double click the cell to make a change.
(2) Use the Preliminary Reference Weights as a start and guide, and interactively develop the Adjusted Weights so that calc. displacement and calc. weight match within 2%, and calc. LCB' and calc. LCG' match within 0.5% LWL.

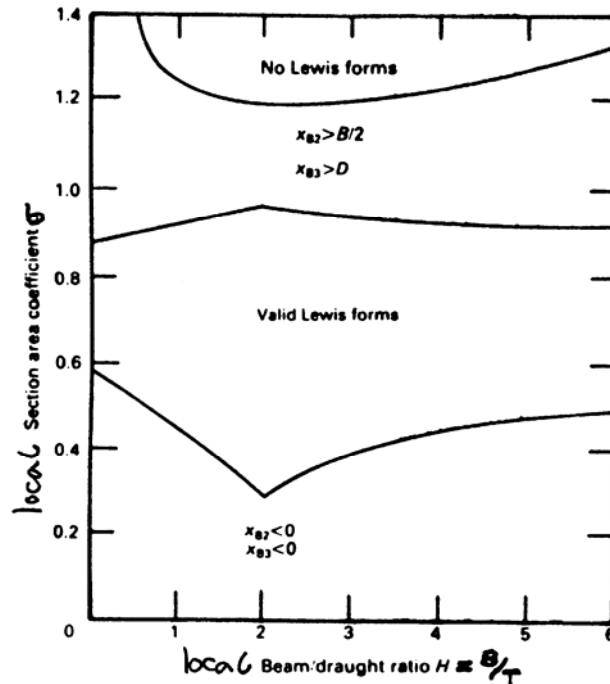
	Sectional			Prelim.		
Station	Beam	Area	Draft	Local	Reference	Adjusted
	(meters)	(meters ²)	(meters)	Ca	Weights	Weights
					(tonnes)	(tonnes)
0	0.000	0.000	0.000		19.995	19.995
1	3.426	3.711	2.500	0.433	39.990	35.000
2	6.107	10.115	2.500	0.662	39.990	38.000
3	7.920	16.389	2.500	0.828	59.984	60.000
4	8.897	20.740	2.500	0.932	59.984	60.000
5	9.180	22.261	2.500	0.970	59.984	60.000
6	8.968	20.799	2.500	0.928	59.984	60.000
7	8.469	16.809	2.500	0.794	59.984	60.000
8	7.851	11.220	2.500	0.572	39.990	47.000
9	7.190	5.296	1.500	0.491	39.990	39.990
10	6.426	0.500	0.130	0.599	19.995	19.995

calc. Cwp	0.776					
calc. displ.	499.87	tonnes		calc. weight	499.98	tonnes
calc. LCB'	-0.92	%LWL, +fwd		calc. LCG'	-0.94	%LWL, +fwd

Note: The SPP rectangular integration algorithm is used for calculation.

Since SPP uses the Lewis Form complex variable transformation to model the hull sections, the user needs to be cognizant of the limits of this transformation. The hull section is transformed to a half circle based upon the local section area coefficient and the local beam-draft ratio. The spreadsheet shows the local section area coefficient for reference. This should vary smoothly along the length and provide a valid Lewis Form as shown in the central band in the figure below. There is not limit to the permissible local beam-draft ratio but only a limited range of the local section area coefficient for which a reasonable Lewis Form is possible.

The spreadsheet also assists the user in developing a discrete weight distribution that provides the required weight~displacement and LCG~LCB. The spreadsheet shows a preliminary discrete weight distribution with the correct weight, but LCG = 0. This can be used for reference while interactively developing the final weight distribution in the next column. The user can use the reference distribution and the calculated LCB to develop a final discrete weight distribution that reflects the details of the particular ship design and provides both the required total weight and LCG. The white cells in the spreadsheet can have data entered or edited by the user by double clicking on the cell to be changed and entering the desired data when prompted.



2. Hull design from calculated data option:

In this second option, a spreadsheet appears which allows the user to input the Design Waterline Curve and the Sectional Area Curve using data obtained from the Hydrostatics analysis of the detailed hull NURB surface or the resulting hull offsets. The development of the discrete weight distribution is the same as in the Hull design from parameters option above.

Calculated Hull Data

Instruction:

- (1) Enter or change data in the columns with white background. To enter or change data, double click the cell.
- (2) Enter beams, sectional areas, and drafts from HYDRO, or other programs.
- (3) Use the Preliminary Reference Weights as a start and guide and work interactively with the Adjust Weights so that calc. displacement and calc. weight match within 2%, and calc. LCB' and calc. LCG' match within 0.5% LWL.
- (4) The SPP rectangular integration algorithm is used for all calculations so results may vary slightly from the Simpson's Rule integration results from HYDRO, or other similar programs.

Hull Data Identifier:

Cancel OK

Hull Design:

		Sectional			Prelim.	
Station	Beam	Area	Draft	Local	Reference	Adjusted
	(meters)	(meters ²)	(meters)	Ca	Weights (tonnes)	Weights (tonnes)
0	0.000	0.000	0.000		19.995	19.995
1	3.426	3.711	2.500	0.433	39.990	33.000
2	6.120	10.115	2.500	0.661	39.990	39.990
3	7.920	16.389	2.500	0.828	59.985	59.985
4	8.897	20.740	2.500	0.932	59.985	59.985
5	9.180	22.262	2.500	0.970	59.985	59.985
6	8.968	20.799	2.500	0.928	59.985	59.985
7	8.469	16.809	2.500	0.794	59.985	59.985
8	7.851	11.220	2.500	0.572	39.990	47.000
9	7.190	5.296	1.300	0.567	39.990	39.990
10	6.426	0.500	0.130	0.599	19.995	19.995

calc. Cwp	0.776				
calc. displ.	499.88	tonnes	calc. weight	499.90	tonnes
calc. LCB'	-0.92	%LWL: +fwd	calc. LCG'	-0.98	%LWL: +fwd

Operating Conditions obtains the vessel speed and the ship heading angle relative to the wave heading.

V_K ship speed in knots
 THETA_w ship heading relative to waves in degrees (following seas 0, beam seas 90, head seas 180)

Seaway Characteristics provides the selection from among the three available seaway spectra models: Pierson-Moskowitz spectrum, ISSC two parameter spectrum, or JONSWAP fetch-limited spectrum. It then obtains the parameters needed to define the selected spectrum.

V_w wind speed in knots, for Pierson-Moskowitz spectrum or JONSWAP fetch-limited spectrum
 FETCH wind fetch in kilometers, for JONSWAP fetch-limited spectrum
 H_s significant wave height in meters, for ISSC two parameter spectrum
 T_1 characteristic wave period in seconds, for ISSC two parameter spectrum

RUNNING SPP:

When the input process is completed, the program can be run through the **Analysis** menu or by selecting the **RunSPP** button. This will produce a final run input window where the user must specify a lower limit and an upper limit for the truncation of the numerical integrations of the input and output spectra needed to obtain the RMS values. The user can also enter a run specific identifier at this point, if desired.

Run SPP ...

Specify Frequency Range:

Lower limit (> 0.03): 0.3 radius/second

Upper limit (< 7.10): 1.8 radius/second

Note: The limits are to be used for numerical integration of spectrum over frequency.

Note: If you want to check all your inputs before running SPP, please click "Verify" button.

Verify

Cancel

OK

Run Identifier: Test Run for Manual

SWL	lower limit of frequency range to be considered in rad/s (>0.03; about 0.3 recommended, but verify this for output of interest)
BWL	upper limit of frequency range to be considered in rad/s (<7.10; about 1.5 recommended, but verify this for output of interest)

The numerical integrals of the input and output spectra are approximated by rectangular integration with a fixed number of ordinates (19) evenly spaced from SWL to BWL. For acceptable numerical accuracy, therefore, it is desirable to have BWL - SWL as small as possible to obtain good resolution of the spectral peak, provided the errors introduced by the truncation of the tails (below SWL and above BWL) are negligible.

CAUTIONS:

- 1. The program may have problems with extremely flat, transom stern sections. In some cases it has been necessary to artificially increase the area of Station 10 up to a local area coefficient of about 0.6 to obtain numerical results. These small changes should have little effect on the overall, total ship results.*

2. The SCORES documentation (1) states the following: "Errors in the calculation of the two-dimensional properties will be self explanatory. However, if an error is found in the energy balance check on the results of the two-dimensional lateral motion calculation the message is printed, but the computations proceed. It has usually been found that such errors in the energy balance check have little influence on the calculated two-dimensional properties." This is not a fatal error so the computations are allowed to proceed and the results are considered to have engineering validity.

3. Hydrodynamic coefficients used in the heave, pitch, and vertical bending moment computations involve division by the encounter frequency which is shown on the program output for reference. In some circumstances in following seas, the encounter frequency passes through zero at a specific value of wave frequency. At these frequencies, the output spectrum ordinates may have very large, physically meaningless values which produce very large response values. This is usually recognizable when the response at a specific value of θ_w is clearly inconsistent with the values at adjacent wave angles. The results at these wave angles should be ignored as physically unreliable.

OUTPUT:

A sample output report resulting from the input from the above windows using the **Hull design from parameters** option follows.

Univeristy of Michigan
Department of Naval Architecture and Marine Engineering

Seakeeping Prediction Program (SPP-1.5) by M.G. Parsons

Reference: Raff, A. I., "Program SCORES - Ship Structural Response
in Waves", Ship Structures Committee Report SSC-230, 1972

Hull Data Identification: Test Hull Design

Run Identification: Test Run for Manual

Input Verification:

Length of Waterline LWL (m)	=	38.00
Vessel Displacement (tonnes)	=	499.9
Vertical Center of Gravity VCG (m)	=	3.50
Roll Radius of Gyration k11 (m)	=	3.80
Fraction of Critical Roll Damping	=	0.1300
Ship Speed (knots)	=	12.00
Ship Heading Relative to Waves (deg)	=	60.00
Water Type	=	Salt@15C
Water Density Rho (kg/m^3)	=	1025.87
ISSC Two Parameter Spectrum Excitation		
Significant Wave Height (m)	=	2.25
Characteristic Wave Period (s)	=	10.00
Lower Freq. Integration Limit (rad/s)	=	0.30
Upper Freq. Integration Limit (rad/s)	=	1.80

Sta.	Beam[m]	Area[m^2]	Draft[m]	Weight[t]
0	0.00	0.00	0.00	20.0
1	3.43	3.71	2.50	35.0
2	6.11	10.11	2.50	38.0
3	7.92	16.39	2.50	60.0
4	8.90	20.74	2.50	60.0
5	9.18	22.26	2.50	60.0
6	8.97	20.80	2.50	60.0
7	8.47	16.81	2.50	60.0
8	7.85	11.22	2.50	47.0
9	7.19	5.30	1.50	40.0
10	6.43	0.50	0.13	20.0

Univeristy of Michigan
Department of Naval Architecture and Marine Engineering

Seakeeping Prediction Program (SPP-1.5) by M.G. Parsons

Reference: Raff, A. I., "Program SCORES - Ship Structural Response
in Waves", Ship Structures Committee Report SSC-230, 1972

Hull Data Identification: Test Hull Design

Run Identification: Test Run for Manual

Motion Natural Frequencies and Periods:

Heave Natural Frequency = 1.551 rad/s Heave Natural Period = 4.05 sec.
Pitch Natural Frequency = 1.563 rad/s Pitch Natural Period = 4.02 sec.
Roll Natural Frequency = 0.704 rad/s Roll Natural Period = 8.93 sec.

Roll Damping Results:

Roll Wave Damping = 0.910E+01
Added Viscous Roll Damping = 0.154E+03

Seakeeping Response Results:

Ship Speed = 12.0 knots = 6.17 m/s
Wave Angle [with Head Seas 180 deg.] = 60.0 deg.
ISSC Two Parameter Spectrum - Sign.Height = 2.25 m Char. Period = 10.00 s

Nondimensional Motion Response Amplitude Operators:

Wave Freq. r/s	Encount. Freq. r/s	Wave Length m	Heave Ampl. n.d.	Heave Phase deg.	Pitch Ampl. n.d.	Pitch Phase deg.	Roll Ampl. n.d.	Roll Phase deg.
0.300	0.272	684.6	0.997E+00	179.9	0.514E+00	-89.5	0.972E+00	-92.5
0.383	0.337	419.3	0.993E+00	179.8	0.512E+00	-88.8	0.104E+01	-92.7
0.467	0.398	282.9	0.987E+00	179.6	0.509E+00	-87.9	0.112E+01	-93.1
0.550	0.455	203.7	0.977E+00	179.3	0.504E+00	-86.8	0.124E+01	-93.8
0.633	0.507	153.6	0.963E+00	179.0	0.497E+00	-85.4	0.138E+01	-95.0
0.717	0.555	120.0	0.943E+00	178.6	0.487E+00	-83.8	0.156E+01	-97.2
0.800	0.599	96.3	0.918E+00	178.1	0.474E+00	-81.9	0.179E+01	-101.0
0.883	0.638	79.0	0.886E+00	177.5	0.458E+00	-79.7	0.203E+01	-107.2
0.967	0.673	65.9	0.846E+00	176.9	0.438E+00	-77.2	0.224E+01	-116.2
1.050	0.703	55.9	0.799E+00	176.1	0.414E+00	-74.4	0.231E+01	-127.2
1.133	0.729	48.0	0.744E+00	175.3	0.386E+00	-71.2	0.217E+01	-137.5
1.217	0.751	41.6	0.682E+00	174.3	0.354E+00	-67.6	0.190E+01	-144.3
1.300	0.768	36.5	0.614E+00	173.1	0.318E+00	-63.7	0.161E+01	-147.2
1.383	0.781	32.2	0.539E+00	171.6	0.279E+00	-59.4	0.135E+01	-146.5
1.467	0.790	28.6	0.459E+00	169.7	0.236E+00	-54.6	0.112E+01	-142.6
1.550	0.794	25.6	0.376E+00	167.0	0.192E+00	-49.3	0.928E+00	-135.7
1.633	0.794	23.1	0.293E+00	162.6	0.147E+00	-43.3	0.759E+00	-125.3
1.717	0.789	20.9	0.214E+00	154.8	0.103E+00	-36.4	0.610E+00	-110.4
1.800	0.780	19.0	0.148E+00	139.6	0.613E-01	-27.6	0.485E+00	-88.7

Seakeeping Prediction Program (SPP-1.5) by M.G. Parsons

Run Identification: Test Run for Manual

Nondimensional Moment Response Amplitude Operators:

Wave Freq. r/s	Encount. Freq. r/s	Wave Length m	Vertical Ampl. n.d.	Moment Phase deg.	Trans. Ampl. n.d.	Moment Phase deg.	Tors. Ampl. n.d.	Moment Phase deg.
0.300	0.272	684.6	0.308E-03	173.7	0.399E-03	88.3	0.195E-04	-138.2
0.383	0.337	419.3	0.449E-03	171.3	0.645E-03	87.9	0.473E-04	-135.3
0.467	0.398	282.9	0.580E-03	166.7	0.938E-03	87.4	0.990E-04	-133.2
0.550	0.455	203.7	0.695E-03	158.5	0.127E-02	86.9	0.188E-03	-131.6
0.633	0.507	153.6	0.813E-03	145.3	0.164E-02	86.3	0.333E-03	-130.9
0.717	0.555	120.0	0.101E-02	127.7	0.205E-02	85.5	0.563E-03	-131.1
0.800	0.599	96.3	0.139E-02	110.3	0.249E-02	84.5	0.913E-03	-132.9
0.883	0.638	79.0	0.204E-02	97.8	0.294E-02	83.4	0.142E-02	-137.0
0.967	0.673	65.9	0.300E-02	90.3	0.341E-02	82.4	0.208E-02	-143.8
1.050	0.703	55.9	0.425E-02	86.6	0.392E-02	82.0	0.276E-02	-152.4
1.133	0.729	48.0	0.581E-02	85.2	0.450E-02	82.0	0.330E-02	-160.4
1.217	0.751	41.6	0.762E-02	85.4	0.523E-02	81.8	0.363E-02	-165.0
1.300	0.768	36.5	0.961E-02	86.7	0.604E-02	81.1	0.383E-02	-166.0
1.383	0.781	32.2	0.117E-01	88.8	0.694E-02	80.2	0.396E-02	-164.2
1.467	0.790	28.6	0.137E-01	91.4	0.788E-02	79.2	0.405E-02	-160.3
1.550	0.794	25.6	0.154E-01	94.5	0.879E-02	78.2	0.409E-02	-154.9
1.633	0.794	23.1	0.167E-01	98.1	0.963E-02	77.5	0.405E-02	-148.2
1.717	0.789	20.9	0.173E-01	102.0	0.103E-01	77.1	0.389E-02	-140.4
1.800	0.780	19.0	0.170E-01	106.5	0.107E-01	77.1	0.352E-02	-131.1

Wave Input and Response Amplitude Spectra:

Freq. r/s	Wave Amp. m^2s	Heave m^2s	Pitch deg.^2s	Roll deg.^2s	Vert. Mom. (t-m)^2s	Lat. Mom. (t-m)^2s	Tors. Mom. (t-m)^2s
0.300	0.007	0.007	0.000	0.002	0.120E+00	0.201E+00	0.481E-03
0.383	0.426	0.420	0.082	0.338	0.159E+02	0.328E+02	0.176E+00
0.467	0.918	0.894	0.385	1.878	0.572E+02	0.149E+03	0.166E+01
0.550	0.818	0.780	0.648	3.900	0.730E+02	0.245E+03	0.533E+01
0.633	0.560	0.519	0.759	5.861	0.685E+02	0.280E+03	0.115E+02
0.717	0.357	0.318	0.762	7.861	0.672E+02	0.278E+03	0.209E+02
0.800	0.226	0.191	0.711	10.101	0.807E+02	0.259E+03	0.349E+02
0.883	0.146	0.114	0.635	12.548	0.112E+03	0.233E+03	0.545E+02
0.967	0.096	0.069	0.550	14.437	0.160E+03	0.207E+03	0.766E+02
1.050	0.065	0.042	0.463	14.373	0.217E+03	0.184E+03	0.917E+02
1.133	0.045	0.025	0.379	11.955	0.281E+03	0.169E+03	0.908E+02
1.217	0.032	0.015	0.300	8.645	0.343E+03	0.162E+03	0.778E+02
1.300	0.023	0.009	0.228	5.856	0.395E+03	0.156E+03	0.625E+02
1.383	0.017	0.005	0.165	3.870	0.430E+03	0.152E+03	0.493E+02
1.467	0.013	0.003	0.112	2.536	0.442E+03	0.146E+03	0.387E+02
1.550	0.010	0.001	0.070	1.646	0.427E+03	0.139E+03	0.301E+02
1.633	0.007	0.001	0.039	1.047	0.387E+03	0.128E+03	0.228E+02
1.717	0.006	0.000	0.018	0.646	0.324E+03	0.115E+03	0.163E+02
1.800	0.005	0.000	0.006	0.390	0.247E+03	0.980E+02	0.106E+02

Wave Input and Response Amplitude Statistics:

	m	m	deg.	deg.	t-m	t-m	t-m
R.M.S.	0.561	0.533	0.725	2.996	0.183E+02	0.160E+02	0.759E+01
Ave.	0.701	0.666	0.906	3.745	0.228E+02	0.200E+02	0.948E+01
Signif.	1.121	1.066	1.450	5.991	0.365E+02	0.321E+02	0.152E+02
Ave 1/10	1.430	1.359	1.849	7.639	0.466E+02	0.409E+02	0.193E+02
Design Value with N=1000 and alpha=0.01					0.877E+02	0.770E+02	0.364E+02



GEAR SIZING PROGRAM (GSP1.0)

M. G. Parsons, January, 1991

REFERENCES:

1. Balukjian, H., "A Computerized Method for the Preliminary Design of Main Reduction Gears," *Naval Engineers Journal*, February, 1972.
2. Mowers, G. P., "Reduction Gears," Ch. IX in *Marine Engineering*, Harrington, R. L., (Ed.), SNAME, Jersey City, N. J., 1992.
3. Parsons, M. G., "NA330 Class Notes for Ship Power Systems I," September, 1977.

METHODS:

This program implements the gear sizing and arrangement method presented by Balukjian (1), Mowers (2) and Parsons (3). The user should consult Balukjian and Mowers prior to using the program.

Based on the input, the program calculates the following gear characteristics:

- total reduction ratio.
- first reduction ratio, if a double reduction gear.
- diameter of the pinion for the first reduction.
- diameter of the gear for the first reduction.
- diameter of the pinion for the second reduction, if a double reduction gear.
- diameter of the gear for the second reduction, if a double reduction gear.
- estimate of the total gear weight.
- estimate of the overall gear box length.
- shaft separation (vertical or horizontal or both, depending on the case analyzed).
- estimate of the overall breadth of the gear box.
- estimate of the overall height of the gear box.

The user should compare the dimensions with the appropriate drawing in the Appendix. For example, if the user has input data for two engines, one power path per engine, and a single reduction gear, then the appropriate drawing is Case 5. Note that in this case the only available offset is horizontal.

INPUT:

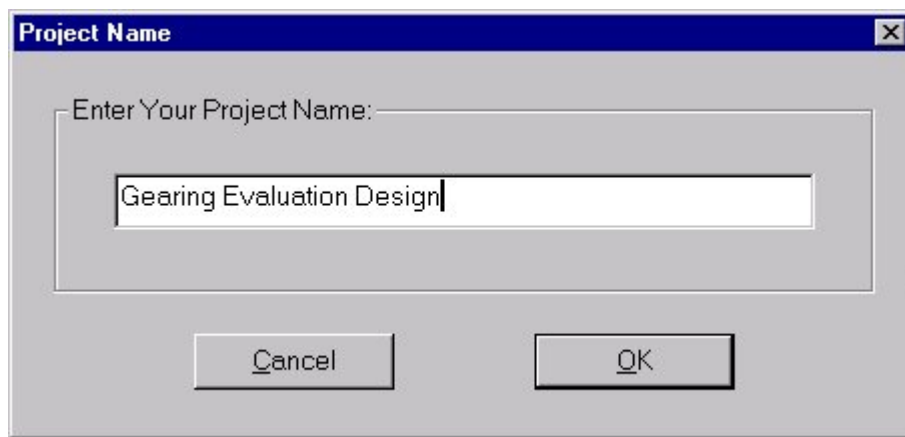
The input to GSP is through a series of three windows within the menu **Input** as follows:

Input

Project Name
Gear Functions
Gear Details

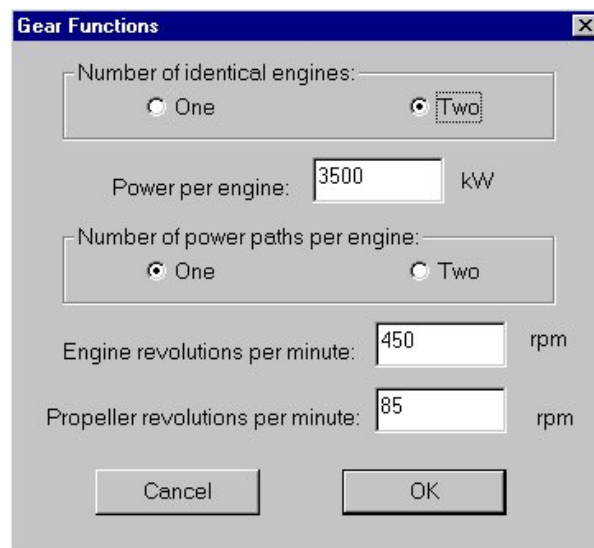
Each of these windows must be completed before an analysis can be performed. The completed input data can be saved using the **Save** button for later re-use.

Project Name provides a location for the general identification of the project being analyzed.



The screenshot shows a dialog box titled "Project Name" with a close button (X) in the top right corner. Inside the dialog, there is a label "Enter Your Project Name:" followed by a text input field. The input field contains the text "Gearing Evaluation Design". Below the input field, there are two buttons: "Cancel" and "OK".

Gear Functions obtains the basic functionality of the reduction gear including number of identical input engines (1 or 2), number of power paths per engine (1 or 2), input power per engine, engine input RPM, and propeller output RPM.



The screenshot shows a dialog box titled "Gear Functions" with a close button (X) in the top right corner. Inside the dialog, there are several input fields and radio buttons. The first section is "Number of identical engines:" with two radio buttons: "One" and "Two". The "Two" radio button is selected. Below this is a text input field for "Power per engine:" containing the value "3500" and the unit "kW". The next section is "Number of power paths per engine:" with two radio buttons: "One" and "Two". The "One" radio button is selected. Below this is a text input field for "Engine revolutions per minute:" containing the value "450" and the unit "rpm". The final section is a text input field for "Propeller revolutions per minute:" containing the value "85" and the unit "rpm". At the bottom of the dialog, there are two buttons: "Cancel" and "OK".

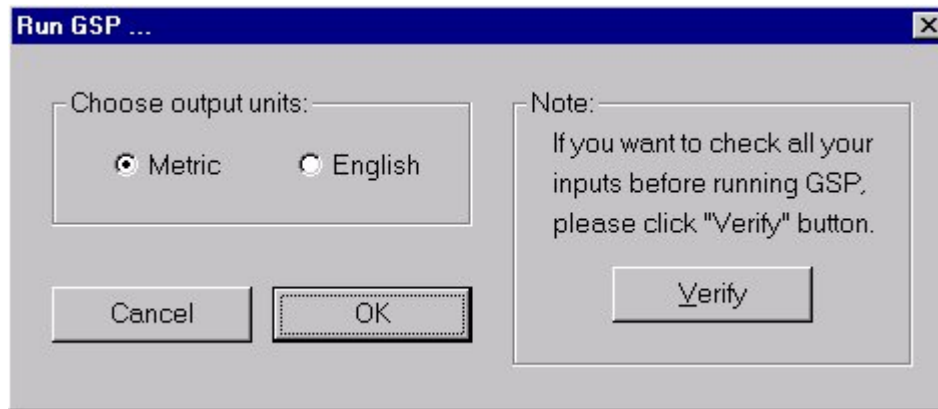
kW input power per engine in kilowatts
 RPM_E engine input revolutions per minute
 RPM_P propeller output revolutions per minute

Gear Details obtains the requirement for a single or double reduction gear and then the appropriate parameters for the gear design selected.

- R_2 second reduction ratio if double reduction gear; a first approximation for R_2 can be taken as $R_2 = 0.2 \cdot (R_{OA}) + 1$, where $R_{OA} = \text{RPM}_E / \text{RPM}_P$
 I_{VH} vertical or horizontal input/output offset configuration when a choice; two engine, double reduction gears have a default 45 degree offset
 K_1 K-factor for the first (high speed) reduction - a dimensional variable with units lbf/in-in; recommended values:
 - 140 commercial, 225 Navy for the first reduction of double reduction gears
 - 110 commercial, 175 Navy for single reduction gears K_2 K-factor for the second reduction of double reduction gears recommended values:
 - 110 commercial, 175 Navy RLD_1 length-diameter L/D ratio for the first reduction pinion recommended value 2.0 or less
 RLD_2 length-diameter L/D ratio for the second reduction pinion of double reduction gears recommended value 2.0 or less

RUNNING GSP:

When the input process is completed, the program can be run through the **Analysis** menu or by selecting the **RunGSP** button. This will produce a final run input window where the user must specify the output units in metric or English.



OUTPUT:

A sample output report resulting from the input from the above windows follows:

The University of Michigan
Department of Naval Architecture and Marine Engineering

Gear Sizing Program (GSP1.0) by M. G. Parsons - January, 1991

Source: Balukjian, H., "A Computerized Method for the
Preliminary Design of Main Reduction Gears,"
Naval Engineers Journal, February, 1972.

Input Verification:

Power per Engine (kW)	=	3500.00
Number of Identical Engines	=	2
Total Power (kW)	=	7000.00
Total Horsepower (BrHP)	=	9387.15
Number of Power Paths per Engine	=	1
Engine Revolutions per Minute (rpm)	=	450.00
Propeller Revolutions per Minute (rpm)	=	85.00
Reduction Gear Type	=	single reduction
Input/Output Offset Configuration	=	horizontal
K-Factor	=	110.00
Pinion L/D Ratio	=	2.00
Results units	=	metric

Calculations Results:

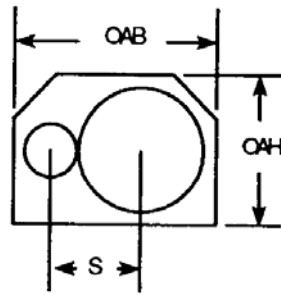
Total Reduction Ratio	=	5.29
Pinion Diameter (m)	=	0.49
Gear Diameter (m)	=	2.59
Total Gear Weight (tonnes)	=	35.26
Overall Gear Length (m)	=	2.69
Shaft Separation (Horizontal) (m)	=	3.07
Overall Breadth of the Gear Box (m)	=	4.27
Overall Height of the Gear Box (m)	=	3.10

Appendix: Reduction Gear Geometries

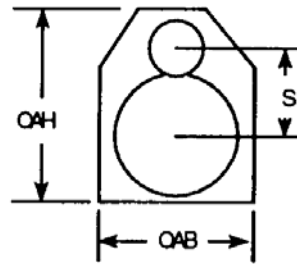
OAB=overall breadth; OAH=overall height;

S, HS, and VS=input/output or input/input shaft (horizontal/vertical) separation

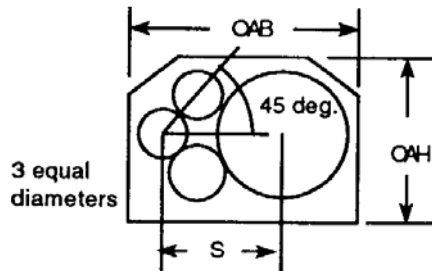
Case 1: one engine, single reduction, horizontal offset, one power path



Case 2: one engine, single reduction, vertical offset, one power path



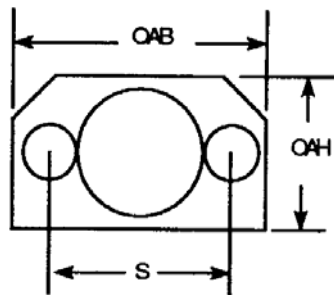
Case 3: one engine, single reduction, horizontal offset, two power paths



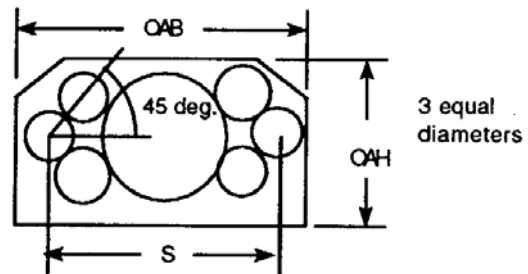
Case 4: one engine, single reduction, vertical offset, two power paths

As in Case 3, but rotated. Dimensions as in Case 2.

Case 5: two engines, single reduction, horizontal offset, one power path

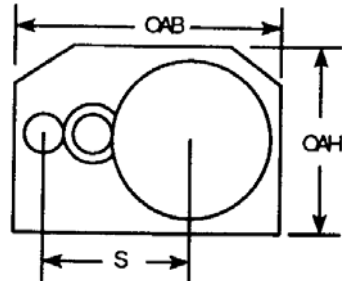


Case 6: two engines, single reduction, horizontal offset, two power paths

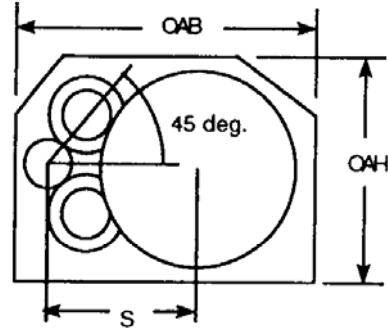


Appendix: Reduction Gear Geometries (cont.)

Case 7: one engine, double reduction, horizontal offset, one power path



Case 8: one engine, double reduction, horizontal offset, two power paths



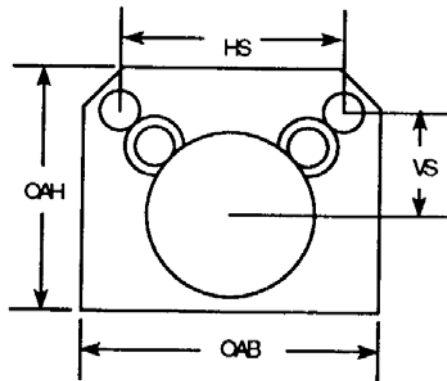
Case 9: one engine, double reduction, vertical offset, one power path

As in Case 7, but rotated. Dimensions as in Case 2.

Case 10: one engine, double reduction, vertical offset, two power paths

As in Case 8, but rotated. Dimensions as in Case 2.

Case 11: two engines, double reduction, 45 degree offset, one power path



Case 12: two engines, double reduction, 45 degree offset, two power paths

